Machine Learning & Deep Learning

Lecture Notes

Pedro Watuhã dos Santos Guerra

2026

A self-contained graduate text covering statistical learning theory, regularisation, tree ensembles, neural networks, variational and generative models, causal inference, and double machine learning.

Acknowledgements

The author gratefully acknowledges the support of **CAPES** (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior), Brazil, for funding this research and the development of these lecture materials under the CAPES Graduate Fellowship Program.

These notes grew from lectures delivered in 2026 and draw on several decades of accumulated wisdom in statistical learning theory and machine learning. The author thanks the many researchers whose foundational work made this synthesis possible.

Preface

These notes cover eight lectures on machine learning and deep learning from a unified statistical perspective. The organizing philosophy is consistent throughout: every algorithm has a mathematical reason for working, and every theorem has an algorithmic consequence.

Scope and arc. The eight lectures form a natural progression. We begin with the foundations of *generalization theory* (Lecture 1): how can we hope to do well on unseen data? Lectures 2–4 develop the three canonical families of prediction methods — regularized linear models, tree-based ensembles, and neural networks — each with their theoretical guarantees and known rates. Lectures 5–6 pivot to *generative modeling*: we shift from predicting a label to learning the full data distribution, treating posterior inference as optimization (variational methods, VAEs) and generation as deterministic transport (flow matching and diffusion). Lectures 7–8 close with *causal inference*: we ask not what the data says but what would happen if we intervened, culminating in Double Machine Learning as the principled bridge between ML flexibility and statistical inference.

Style. Each topic is developed from motivation through formalism to proof. Historical notes situate the ideas in context; examples and figures illustrate the theory; implementation insights connect the mathematics to practice. Proofs are given in full whenever they illuminate the method, and sketched whenever the full argument would require substantial measure-theoretic prerequisites beyond the course scope.

Prerequisites. Graduate-level probability and statistics; linear algebra (spectral decomposition, matrix calculus); real analysis at the level of sequences and continuity; working knowledge of Python (NumPy, scikit-learn, PyTorch).

Notation.

- $\mathbb{E}$: expectation; $\mathbb{P}$: probability; $\mathbb{E}_n f = \frac{1}{n}\sum_{i=1}^n f(x_i)$: empirical mean.
- $O_{\mathbb{P}}(\cdot)$, $o_{\mathbb{P}}(\cdot)$: stochastic big-O and little-o (convergence in probability).
- $\|\cdot\|_p$: ℓ^p norm; $\|\cdot\| = \|\cdot\|_2$ unless stated.
- $[n] = \{1, \ldots, n\}$; $\operatorname{sign}(z)$: sign of z; $(z)_+ = \max(z, 0)$.
- Boldface lowercase $\boldsymbol{x}$ for vectors; uppercase X for random variables.

Contents

Acknowledgements **i**

Preface **ii**

1 Statistical Learning Theory **1**
1.1 The Supervised Learning Problem . . . 1
1.1.1 The Bayes-Optimal Predictor . . . 2
1.1.2 Realizability vs. Agnostic Setting . . . 3
1.2 Concentration Inequalities . . . 3
1.2.1 Finite Hypothesis Classes: The Union Bound Argument . . . 5
1.2.2 McDiarmid's Inequality (Bounded Differences) . . . 6
1.3 Uniform Convergence and Rademacher Complexity . . . 6
1.4 VC Dimension and Sauer's Lemma . . . 8
1.5 The Bias–Variance Decomposition . . . 9
1.6 No Free Lunch and the Need for Priors . . . 11

2 Regularisation **12**
2.1 The Linear Model and Its Limits . . . 12
2.1.1 The Parametric Rate . . . 13
2.1.2 Minimax Lower Bounds . . . 13
2.2 Ridge Regression . . . 14
2.2.1 Singular Value Analysis of Ridge . . . 14
2.2.2 Bias-Variance Analysis of Ridge . . . 15
2.2.3 Bayesian Interpretation . . . 15
2.2.4 Oracle Inequality for Ridge . . . 15
2.2.5 Support Recovery Conditions for LASSO . . . 16
2.3 The LASSO . . . 16
2.3.1 Coordinate Descent for LASSO . . . 17
2.3.2 Statistical Theory: Sparsity and Rates . . . 18
2.4 Model Selection: Cross-Validation and the Regularisation Path . . . 19

3 Decision Trees and Random Forests **20**
3.1 Decision Trees . . . 20
3.1.1 Why Trees? . . . 20
3.1.2 The CART Algorithm . . . 21
3.1.3 The Bias-Variance Tradeoff in Trees . . . 22
3.1.4 Cost-Complexity Pruning . . . 23

3.1.5 Minimax Rate for Trees . . . 23
3.2 Ensemble Methods . . . 23
3.2.1 The Variance Reduction Argument . . . 23
3.2.2 Bagging . . . 24
3.2.3 Out-of-Bag Error Estimation . . . 24
3.3 Random Forests . . . 24
3.3.1 Variable Importance . . . 25
3.3.2 Causal Forests . . . 25
3.3.3 Consistency Results . . . 26
3.4 Benign Overfitting and Double Descent . . . 26

4 Neural Networks and Approximation Theory 28
4.1 The Perceptron . . . 28
4.2 Multi-Layer Perceptrons . . . 29
4.2.1 Activation Functions . . . 29
4.3 Backpropagation . . . 30
4.3.1 The Optimization Problem . . . 30
4.3.2 The Forward Pass and Error Signals . . . 30
4.3.3 The Backpropagation Recursion . . . 30
4.3.4 Vanishing and Exploding Gradients . . . 31
4.4 Approximation Theory . . . 31
4.4.1 Universal Approximation . . . 31
4.4.2 Barron's Theorem: Quantitative, Dimension-Free Approximation . . . 32
4.5 The Lazy / NTK Regime and Benign Overfitting in Neural Networks . . . 33
4.5.1 The Linear / Lazy Regime . . . 34
4.5.2 Stochastic Gradient Descent . . . 34

5 Variational Inference, VAEs, and Langevin Dynamics 36
5.1 From Prediction to Generative Modeling . . . 36
5.2 Information Theory: Entropy and KL Divergence . . . 37
5.2.1 KL is Not Symmetric: Forward vs. Reverse KL . . . 37
5.3 The Evidence Lower Bound (ELBO) . . . 38
5.4 Coordinate Ascent Variational Inference (CAVI) . . . 39
5.5 Variational Autoencoders . . . 40
5.5.1 The Reparameterization Trick . . . 41
5.5.2 Geometry of the VAE Latent Space . . . 41
5.6 Langevin Dynamics . . . 42

6 Diffusion Models and Flow Matching 43
6.1 Distances Between Distributions . . . 43
6.2 Denoising Diffusion Probabilistic Models (DDPM) . . . 44
6.2.1 From VAE to DDPM . . . 44
6.2.2 The Forward Process . . . 44
6.2.3 The Reverse Process . . . 45
6.2.4 Training: The ELBO and Noise Prediction . . . 45
6.3 Score Matching . . . 46
6.3.1 The Score Function . . . 46
6.3.2 Score Matching (Hyvärinen, 2005) . . . 46
6.3.3 Denoising Score Matching (Vincent, 2011) . . . 47
6.4 Flow Matching . . . 47
6.4.1 The ODE Perspective . . . 47

6.4.2 Conditional Flow Matching (CFM) . . . 48
6.4.3 Gaussian / Linear Paths . . . 48
6.4.4 The Unifying Framework: Score SDE and PF-ODE . . . 49

7 Causal Inference **50**
7.1 Why Causation Cannot be Reduced to Correlation . . . 50
7.2 Pearl's Ladder of Causation . . . 51
7.3 The Potential Outcomes Framework (Rubin, 1974) . . . 51
7.3.1 Estimands . . . 51
7.3.2 Identification under Ignorability . . . 52
7.4 Structural Causal Models and DAGs . . . 52
7.4.1 The Three Elementary Path Motifs . . . 52
7.5 Instrumental Variables . . . 53
7.5.1 When Backdoor Fails: Hidden Confounders . . . 53
7.5.2 The Wald Estimator and LATE . . . 54
7.5.3 Manski Partial Identification Bounds . . . 55
7.6 Causal Discovery: The PC Algorithm . . . 55

8 Double Machine Learning **57**
8.1 The Problem: High-Dimensional Controls with Valid Inference . . . 57
8.2 The Partially Linear Model . . . 58
8.3 Why Naive ML Fails . . . 58
8.4 Robinson's Decomposition . . . 58
8.4.1 Frisch-Waugh-Lovell (FWL) Theorem . . . 59
8.4.2 Nonparametric Extension: Robinson's Equation . . . 59
8.5 The DML Solution . . . 59
8.5.1 From $a + b$ to $a^* + b^* + c^*$ . . . 59
8.5.2 Ingredient 1: Neyman Orthogonality . . . 60
8.5.3 Ingredient 2: Cross-Fitting . . . 61
8.6 When Are Rate Conditions Achievable? . . . 62
8.7 The Interactive Model and Doubly Robust Estimation . . . 62
8.8 Instrumental Variables in the DML Framework . . . 63
8.8.1 DML-IV Estimator . . . 63
8.8.2 Weak Instruments . . . 64

Bibliography **65**

List of Figures

1.1 Width of the 95% Hoeffding confidence interval around the empirical mean as a function of sample size n. The interval shrinks as $O(n^{-1/2})$. 5

1.2 Train and test MSE as a function of polynomial degree on a synthetic dataset ($n = 200$, 5-fold CV). The gap between train and test error widens sharply for high degrees, illustrating overfitting. 10

2.1 (a) The soft-thresholding operator $S_\lambda(z)$ sets small inputs to exactly zero, unlike hard thresholding which creates discontinuities. (b) As λ increases, LASSO coefficients hit zero at finite λ (sparsity), while Ridge coefficients only approach zero asymptotically. 17

3.1 Feature space partition induced by a depth-4 decision tree on the two-moons dataset. Each region is labeled by its majority class. Axis-aligned boundaries are a limitation — the true boundary is curved. 22

3.2 Squared bias, variance, and total test MSE vs. tree depth on a 1D regression problem. Variance grows monotonically with depth; bias decreases. The optimal depth balances these two. 23

3.3 Double descent curve: test MSE vs. model dimension p for linear regression with $n = 50$ samples. The peak at $p = n$ is the interpolation threshold. For $p \gg n$, the minimum-norm interpolator achieves near-zero bias and declining variance. 27

4.1 Five common activation functions. ReLU is the workhorse of modern deep learning. GELU (used in transformers) smoothly interpolates between a linear function and ReLU. 29

5.1 Forward vs. reverse KL on a bimodal target $p(z|x)$. Forward KL produces a broad, mass-covering approximation. Reverse KL produces a narrow, mode-seeking approximation (mode collapse). Both are unimodal Gaussians, so neither perfectly captures the bimodal structure. 38

6.1 Forward diffusion process: the distribution $q(x_t|x_0)$ for $x_0 = 2$ becomes progressively broader and centered on zero. By $t = T$, the signal is destroyed and $q(x_T|x_0) \approx \mathcal{N}(0, 1)$. 44

6.2 Score function of a bimodal distribution $p(x) = 0.5\mathcal{N}(-2, 1) + 0.5\mathcal{N}(2, 1)$. The score $\nabla_x \log p(x)$ (red) points toward the two modes, with a discontinuity at the saddle between modes. Langevin dynamics following this vector field converges to samples from p. 46

6.3 Gaussian flow matching: 20 straight-line sample paths from $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$ to a 3-component Gaussian mixture. Each path is a linear interpolation $(1-t)x_0 + tx_1$ with constant velocity $x_1 - x_0$. No ODE simulation is needed during training — only the endpoints. 49

7.1 Instrumental variable DAG: Z (instrument) affects treatment T, which affects outcome Y. The unobserved confounder U (dashed) affects both T and Y. Crucially, Z has no direct path to Y and is independent of U. . . . 54

8.1 Sampling distributions of $\sqrt{n}(\hat{\theta} - \theta_0)$ from 500 simulations ($n = 500$, $p = 20$). Left: naive OLS plug-in — severely biased, non-normal. Center: LASSO plug-in — biased, shifted from zero. Right: DML with cross-fitting — centered on zero, well-approximated by $N(0, \Sigma)$. Only DML provides valid inference. 62

Lecture 1

Statistical Learning Theory

Historical Note: The Birth of Statistical Learning Theory

The formal study of when and why learning from data works dates to the 1960s–1980s. Vapnik and Chervonenkis (1971) proved the first uniform convergence results and introduced what we now call VC dimension. Valiant (1984) formalized the PAC (Probably Approximately Correct) learning framework, asking: how many samples suffice to learn a concept class? The Rademacher complexity framework emerged later, in work by Bartlett and Mendelson (2002), providing tighter and more broadly applicable bounds. These ideas form the bedrock of everything that follows.

1.1. The Supervised Learning Problem

Supervised learning asks the following question. We observe n labeled examples

$$\mathcal{D} = \{(x_1, y_1), \ldots, (x_n, y_n)\} \subset \mathcal{X} \times \mathcal{Y}$$

drawn i.i.d. from an unknown joint distribution P_{XY} on $\mathcal{X} \times \mathcal{Y}$. We want to find a function $f : \mathcal{X} \to \mathcal{Y}$ that makes good predictions on *future* data drawn from the same distribution.

The quality of a prediction is measured by a **loss function** $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, \infty)$. Common examples:

- *Squared loss* (regression): $\ell(y, \hat{y}) = (y - \hat{y})^2$.
- *0-1 loss* (classification): $\ell(y, \hat{y}) = \mathbb{1}[y \neq \hat{y}]$.
- *Logistic loss*: $\ell(y, \hat{y}) = \log(1 + e^{-y\hat{y}})$.

Definition: Population and Empirical Risk

The **population risk** (true risk) of f is:

$$R(f) = \mathbb{E}_{(X,Y)\sim P}[\ell(f(X), Y)].$$

Since P is unknown, we work with the **empirical risk** (training error):

$$R_n(f) = \frac{1}{n} \sum_{i=1}^{n} \ell(f(x_i), y_i) = \mathbb{E}_n[\ell(f(X), Y)].$$

The population risk $R(f)$ is the quantity we truly care about — it measures how f will perform on unseen data. The empirical risk $R_n(f)$ is what we can compute, but it is a biased proxy: any function that memorizes the training data will have $R_n(f) = 0$ regardless of its true performance.

Definition: Empirical Risk Minimisation (ERM)

Given a **hypothesis class** $\mathcal{H} \subseteq \{f : \mathcal{X} \to \mathcal{Y}\}$, the ERM learner returns:

$$\widehat{f}_n = \arg\min_{f \in \mathcal{H}} R_n(f).$$

1.1.1. The Bayes-Optimal Predictor

Before analyzing ERM, we should understand the best possible predictor — the **Bayes-optimal** solution — so we know the target.

Bayes-Optimal Classifier (0-1 loss)

Let $\eta(x) = P(Y = 1|X = x)$. The population risk minimizer is the **Bayes classifier**:

$$f^*(x) = \mathbb{1}[\eta(x) \geq \tfrac{1}{2}].$$

Proof

By the law of total expectation:

$$\begin{aligned} R(h) = \mathbb{E}[\mathbb{1}[h(X) \neq Y]] &= \mathbb{E}_X\Big[\mathbb{E}_{Y|X}[\mathbb{1}[h(X) \neq Y] \mid X]\Big] \\ &= \mathbb{E}_X[P(Y \neq h(X)|X)]\,. \end{aligned}$$

If $h(x) = 1$: $P(Y \neq 1|X = x) = 1 - \eta(x)$. If $h(x) = 0$: $P(Y \neq 0|X = x) = \eta(x)$. To minimize pointwise: output 1 when $1 - \eta(x) < \eta(x)$, i.e., $\eta(x) \geq 1/2$. Hence $f^*(x) = \mathbb{1}[\eta(x) \geq 1/2]$, achieving the **Bayes error rate** $R^* = \mathbb{E}[\min(\eta(X), 1-\eta(X))]$. □

Bayes-Optimal Regressor (squared loss)

Under squared loss $\ell(y, \hat{y}) = (y - \hat{y})^2$, the population risk minimizer is the conditional expectation:

$$f^*(x) = \mathbb{E}[Y|X = x].$$

The irreducible error (Bayes risk) is $R^* = \mathbb{E}[\mathrm{Var}(Y|X)]$.

Proof

Write $Y = f^*(X) + \eta$ where $\eta = Y - \mathbb{E}[Y|X]$ satisfies $\mathbb{E}[\eta|X] = 0$. For any h:

$$\begin{aligned} R(h) &= \mathbb{E}[(h(X) - Y)^2] = \mathbb{E}_X \mathbb{E}_{Y|X}[(h(X) - f^*(X) + f^*(X) - Y)^2|X] \\ &= \mathbb{E}_X\left[(h(X) - f^*(X))^2 + 2(h(X) - f^*(X))\underbrace{\mathbb{E}[f^*(X) - Y|X]}_{=0} + \underbrace{\mathbb{E}[(Y - f^*(X))^2|X]}_{\mathrm{Var}(Y|X)}\right] \\ &= \underbrace{\mathbb{E}[(h(X) - f^*(X))^2]}_{\geq 0,\ =0 \text{ iff } h=f^*} + \underbrace{\mathbb{E}[\mathrm{Var}(Y|X)]}_{R^*}. \end{aligned}$$

□

1.1.2. Realizability vs. Agnostic Setting

- **Realizable (ideal) case**: $f^* \in \mathcal{H}$. The approximation error is zero; we only need to control estimation error. This is the "PAC learnable" setting of Valiant (1984).
- **Agnostic (real) case**: f^* may not lie in $\mathcal{H}$. The best in class $f^*_{\mathcal{H}} = \arg\min_{f\in\mathcal{H}} R(f)$ incurs irreducible approximation error $R(f^*_{\mathcal{H}}) - R(f^*) > 0$. Real problems are agnostic — no finite hypothesis class can perfectly approximate f^*.

The agnostic setting is what we encounter in practice, and it is the setting for which Rademacher and VC bounds are most important.

The *generalization gap* $R(\widehat{f}_n) - R_n(\widehat{f}_n)$ captures how much worse $\widehat{f}_n$ performs on new data than on training data. Controlling this gap is the central problem of statistical learning theory.

We decompose the excess risk as:

$$R(\widehat{f}_n) - R(f^*) = \underbrace{R(\widehat{f}_n) - \inf_{f\in\mathcal{H}} R(f)}_{\text{estimation error}} + \underbrace{\inf_{f\in\mathcal{H}} R(f) - R(f^*)}_{\text{approximation error}}, \tag{1.1}$$

where $f^* = \arg\min_f R(f)$ is the Bayes-optimal predictor. The estimation error vanishes as $n \to \infty$ for rich enough $\mathcal{H}$; the approximation error vanishes as $\mathcal{H}$ grows. These pull in opposite directions — richer classes reduce approximation error but increase estimation error. This is the fundamental bias-variance tension.

1.2. Concentration Inequalities

Before analyzing ERM over function classes, we need tools for controlling random fluctuations of averages.

Markov's Inequality

For a non-negative random variable Z and $t > 0$:

$$\mathbb{P}(Z \geq t) \leq \frac{\mathbb{E}[Z]}{t}.$$

Proof

$\mathbb{E}[Z] = \mathbb{E}[Z\mathbb{1}_{Z\geq t}] + \mathbb{E}[Z\mathbb{1}_{Z<t}] \geq \mathbb{E}[Z\mathbb{1}_{Z\geq t}] \geq t\,\mathbb{P}(Z \geq t)$. Divide by t. □

Hoeffding's Inequality

Let $Z_1, \ldots, Z_n$ be independent random variables with $Z_i \in [a_i, b_i]$ almost surely. Let $\bar{Z} = \frac{1}{n}\sum_i Z_i$ and $\mu = \mathbb{E}[\bar{Z}]$. Then for any $t > 0$:

$$\mathbb{P}(\bar{Z} - \mu \geq t) \leq \exp\left(\frac{-2n^2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

For losses bounded in $[0, 1]$: $\mathbb{P}(R_n(f) - R(f) \geq t) \leq e^{-2nt^2}$.

Proof sketch (Hoeffding's Lemma ⇒ Hoeffding's Inequality)

The core ingredient is **Hoeffding's Lemma**: if $Z \in [a, b]$ and $\mathbb{E}[Z] = 0$, then $\mathbb{E}[e^{sZ}] \leq \exp(s^2(b-a)^2/8)$ for all $s > 0$.
Proof of Lemma. By convexity of e^{sz} on $[a, b]$:

$$e^{sz} \leq \frac{b-z}{b-a}e^{sa} + \frac{z-a}{b-a}e^{sb}.$$

Taking expectations with $\mathbb{E}[Z] = 0$: $\mathbb{E}[e^{sZ}] \leq \frac{b}{b-a}e^{sa} - \frac{a}{b-a}e^{sb} =: e^{\phi(s)}$. One verifies $\phi(0) = \phi'(0) = 0$ and $\phi''(s) \leq (b-a)^2/4$, giving $\phi(s) \leq s^2(b-a)^2/8$ by Taylor.
Completing the bound. By Markov applied to $e^{s(\bar{Z}-\mu)}$ with $Z_i' = Z_i - \mathbb{E}[Z_i]$:

$$\begin{aligned}\mathbb{P}(\bar{Z} - \mu \geq t) = \mathbb{P}(e^{s(\bar{Z}-\mu)} \geq e^{st}) &\leq e^{-st}\prod_{i=1}^n \mathbb{E}[e^{sZ_i'/n}] \\ &\leq e^{-st}\prod_{i=1}^n e^{s^2(b_i-a_i)^2/(8n^2)} = e^{-st+s^2\sum_i(b_i-a_i)^2/(8n^2)}.\end{aligned}$$

Optimizing over $s > 0$ (set $s = 4n^2t/\sum_i(b_i - a_i)^2$) gives the result. □

Consequence. For a fixed f and $\ell \in [0, 1]$, with probability $\geq 1 - \delta$:

$$R(f) \leq R_n(f) + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

Figure 1.1 shows how the confidence width $2\sqrt{\log(2/0.05)/(2n)}$ shrinks with sample size.

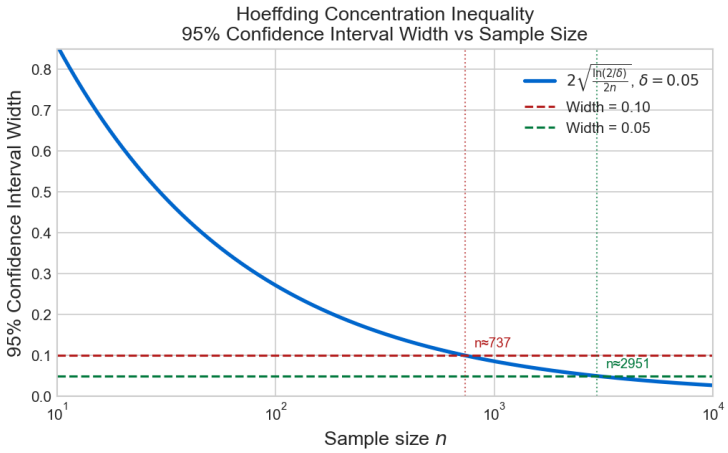


Figure 1.1: Width of the 95% Hoeffding confidence interval around the empirical mean as a function of sample size n. The interval shrinks as $O(n^{-1/2})$.

Hoeffding is for a fixed f

Hoeffding applies to a single hypothesis. For ERM, we simultaneously minimize over all $f \in \mathcal{H}$. The probability bound must hold *uniformly*, which requires stronger tools.

1.2.1. Finite Hypothesis Classes: The Union Bound Argument

As a warm-up to Rademacher complexity, consider the simplest setting: $|\mathcal{H}| < \infty$.

Sample Complexity for Finite Hypothesis Classes

Let $\mathcal{H}$ be finite with $|\mathcal{H}| < \infty$ and $\ell \in [0, 1]$. Then for any $\varepsilon, \delta > 0$, if:

$$n \geq \frac{\log(|\mathcal{H}|/\delta)}{2\varepsilon^2},$$

then $\mathbb{P}[\exists h \in \mathcal{H} : R(h) - R_n(h) \geq \varepsilon] \leq \delta$.

Proof

By Hoeffding for a fixed h: $\mathbb{P}(R(h) - R_n(h) \geq \varepsilon) \leq e^{-2n\varepsilon^2}$. By the **union bound**:

$$\mathbb{P}[\exists h \in \mathcal{H} : R(h) - R_n(h) \geq \varepsilon] \leq \sum_{h \in \mathcal{H}} \mathbb{P}(R(h) - R_n(h) \geq \varepsilon) \leq |\mathcal{H}|\, e^{-2n\varepsilon^2} \leq \delta.$$

Solving: $n \geq \log(|\mathcal{H}|/\delta)/(2\varepsilon^2)$. □

Interpretation. The price of using a richer class is logarithmic in $|\mathcal{H}|$: doubling the number of hypotheses only adds $\log 2/(2\varepsilon^2)$ extra samples. This is why ERM works for finite classes. However, most interesting hypothesis classes (linear functions, neural networks) are uncountably infinite — we need a more refined measure of complexity.

1.2.2. McDiarmid's Inequality (Bounded Differences)

Hoeffding controls averages; what about arbitrary functions of independent random variables?

McDiarmid's Inequality (1989)

Let $Z_1, \ldots, Z_n$ be independent, and let $\Phi : \mathcal{Z}^n \to \mathbb{R}$ satisfy the **bounded differences condition**: for each i,

$$\sup_{z_1,\ldots,z_n,z_i'} |\Phi(z_1, \ldots, z_i, \ldots, z_n) - \Phi(z_1, \ldots, z_i', \ldots, z_n)| \leq c_i.$$

Then for any $\varepsilon > 0$:

$$\mathbb{P}(\Phi(Z) - \mathbb{E}[\Phi(Z)] \geq \varepsilon) \leq \exp\left(\frac{-2\varepsilon^2}{\sum_i c_i^2}\right).$$

Key insight from the hand notes. "The main problem of Hoeffding is that it is all about the mean. McDiarmid bounds the strength of any single data point." This is crucial for the Rademacher bound: $\Phi(S) = \sup_{f\in\mathcal{H}}[R(f) - R_n(f)]$ satisfies bounded differences with $c_i = 2/n$ (changing one point shifts the empirical risk by at most $2/n$), giving $\mathbb{P}(\Phi - \mathbb{E}[\Phi] \geq \varepsilon) \leq e^{-n\varepsilon^2/2}$.

1.3. Uniform Convergence and Rademacher Complexity

The key to analyzing ERM is *uniform* control of the deviation $R(f) - R_n(f)$ over all $f \in \mathcal{H}$.

Definition: Rademacher Complexity

Let $\mathcal{F}$ be a class of real-valued functions on $\mathcal{X}$, and $S = (x_1, \ldots, x_n) \in \mathcal{X}^n$. The **empirical Rademacher complexity** is:

$$\widehat{\mathfrak{R}}_n(\mathcal{F}) = \mathbb{E}_\sigma\left[\sup_{f\in\mathcal{F}} \frac{1}{n}\sum_{i=1}^n \sigma_i f(x_i)\right],$$

where $\sigma_1, \ldots, \sigma_n \overset{\text{i.i.d.}}{\sim} \text{Uniform}\{-1, +1\}$ are **Rademacher random variables**. The population Rademacher complexity is $\mathfrak{R}_n(\mathcal{F}) = \mathbb{E}_S[\widehat{\mathfrak{R}}_n(\mathcal{F})]$.

Intuition. Rademacher variables are pure random labels. The complexity $\widehat{\mathfrak{R}}_n(\mathcal{F})$ measures how well the best function in $\mathcal{F}$ can correlate with completely random noise on the training set S. If $\mathcal{F}$ is rich enough to fit any pattern, including pure noise, then $\widehat{\mathfrak{R}}_n(\mathcal{F})$ is large. Conversely, a restricted class that cannot memorize noise will have small complexity and hence good generalization.

Rademacher Generalization Bound (Bartlett & Mendelson, 2002)

Let $\mathcal{H}$ be a class of functions $f : \mathcal{X} \to [0, 1]$ and $\ell : [0, 1] \times [0, 1] \to [0, 1]$ Lipschitz

with constant L. With probability at least $1-\delta$ over the draw of $S \sim P^n$:

$$\sup_{f\in\mathcal{H}}[R(f) - R_n(f)] \le 2L\,\mathfrak{R}_n(\mathcal{H}) + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

In particular, for the ERM output $\widehat{f}_n$:

$$R(\widehat{f}_n) \le \inf_{f\in\mathcal{H}} R(f) + 4L\,\mathfrak{R}_n(\mathcal{H}) + \sqrt{\frac{2\log(2/\delta)}{n}}.$$

Proof

Step 1: Symmetrisation. Introduce a ghost sample $S' = (x'_1, \dots, x'_n)$ i.i.d. from P. The key symmetrisation lemma states:

$$\mathbb{E}_S\left[\sup_{f\in\mathcal{H}}[R(f) - R_n(f)]\right] \le 2\mathbb{E}_{S,S'}\left[\sup_{f\in\mathcal{H}} \frac{1}{n}\sum_i (R_n^{S'}(f) - R_n^S(f))\right].$$

Proof of symmetrisation. For any fixed f, $R(f) = \mathbb{E}_{S'}[R_n^{S'}(f)]$, so:

$$R(f) - R_n^S(f) = \mathbb{E}_{S'}[R_n^{S'}(f) - R_n^S(f)].$$

Taking supremum over f and expectation over S, and using Jensen's inequality on the concave operation $\mathbb{E}_{S'}[\sup_f \cdot] \ge \sup_f \mathbb{E}_{S'}[\cdot]$ in reverse:

$$\mathbb{E}_S \sup_f [R(f) - R_n^S(f)] \le \mathbb{E}_{S,S'} \sup_f [R_n^{S'}(f) - R_n^S(f)].$$

Step 2: Rademacher randomisation. Since (x_i, x'_i) is a symmetric pair, we can multiply their contributions by $\sigma_i \in \{-1, +1\}$ without changing the distribution:

$$\mathbb{E}_{S,S'} \sup_f \frac{1}{n}\sum_i [\ell(f(x'_i)) - \ell(f(x_i))] = \mathbb{E}_{S,S',\sigma} \sup_f \frac{\sigma_i}{n}\sum_i [\ell(f(x'_i)) - \ell(f(x_i))].$$

Bounding the x' contribution separately gives the Rademacher complexity.
Step 3: McDiarmid's inequality. The function $\Phi(S) = \sup_f[R(f) - R_n^S(f)]$ changes by at most $2/n$ when any single x_i is replaced (since $\ell \in [0,1]$). McDiarmid's inequality then gives the $\sqrt{\log(1/\delta)/(2n)}$ correction term. □

Example: Rademacher Complexity of Linear Functions

Let $\mathcal{H} = \{x \mapsto w^\top x : \|w\|_2 \le B\}$ and suppose $\|x_i\|_2 \le C$ for all i. Then:

$$\widehat{\mathfrak{R}}_n(\mathcal{H}) = \mathbb{E}_\sigma\left[\sup_{\|w\|\le B} \frac{1}{n}\sum_i \sigma_i w^\top x_i\right] = \frac{B}{n}\mathbb{E}_\sigma\left[\left\|\sum_i \sigma_i x_i\right\|\right]$$
$$\le \frac{B}{n}\sqrt{\mathbb{E}_\sigma\left\|\sum_i \sigma_i x_i\right\|^2} = \frac{B}{n}\sqrt{\sum_i \|x_i\|^2} \le \frac{BC}{\sqrt{n}}.$$

The generalization bound for linear classifiers with bounded norm is $O(BC/\sqrt{n})$, independent of the input dimension p.

1.4. VC Dimension and Sauer's Lemma

For binary classification, a combinatorial complexity measure — the VC dimension — gives an alternative route to generalization bounds.

Definition: VC Dimension

A hypothesis class $\mathcal{H}$ of binary functions $\mathcal{X} \to \{0,1\}$ **shatters** a set $C \subset \mathcal{X}$ if for every binary labeling $b : C \to \{0,1\}$, there exists $h \in \mathcal{H}$ with $h(x) = b(x)$ for all $x \in C$. The **Vapnik–Chervonenkis (VC) dimension** is:

$$\mathrm{VC}(\mathcal{H}) = \sup\{|C| : \mathcal{H} \text{ shatters } C\}.$$

Example: VC Dimension of Halfspaces

The class of halfspaces in $\mathbb{R}^p$, $\mathcal{H} = \{\mathrm{sign}(w^\top x + b) : w \in \mathbb{R}^p, b \in \mathbb{R}\}$, has $\mathrm{VC}(\mathcal{H}) = p + 1$.
Lower bound ($\mathrm{VC} \geq p + 1$): Take the standard basis vectors $e_1, \ldots, e_p$ and the origin 0. For any labeling $(b_1, \ldots, b_{p+1}) \in \{0,1\}^{p+1}$, the halfspace $w^\top x + b \geq 0$ with $w = (b_1 - 1/2, \ldots, b_p - 1/2)$ and appropriate b classifies all $p + 1$ points correctly.
Upper bound ($\mathrm{VC} \leq p + 1$): By Radon's theorem, any $p + 2$ points in $\mathbb{R}^p$ can be partitioned into two sets whose convex hulls intersect. Such a configuration cannot be shattered by a halfspace.

Sauer–Shelah Lemma

Let $d = \mathrm{VC}(\mathcal{H})$. The **growth function** $\Pi_{\mathcal{H}}(n) = \max_{S:|S|=n} |\{(h(x_1), \ldots, h(x_n)) : h \in \mathcal{H}\}|$ satisfies:

$$\Pi_{\mathcal{H}}(n) \leq \sum_{i=0}^{d} \binom{n}{i} \leq \left(\frac{en}{d}\right)^d.$$

Proof

By induction on $n + d$. Let $\mathcal{H}_S = \{(h(x_1), \ldots, h(x_n)) : h \in \mathcal{H}\}$ and fix any x_n. Define:

$$\mathcal{H}^0 = \{(b_1, \ldots, b_{n-1}) : (b_1, \ldots, b_{n-1}, 0) \in \mathcal{H}_S\}, \quad \mathcal{H}^1 = \{(b_1, \ldots, b_{n-1}) : (b_1, \ldots, b_{n-1}, 1) \in \mathcal{H}_S\}.$$

Then $|\mathcal{H}_S| = |\mathcal{H}^0 \cup \mathcal{H}^1| = |\mathcal{H}^0| + |\mathcal{H}^1 \setminus \mathcal{H}^0|$. By induction on $\{x_1, \ldots, x_{n-1}\}$: $|\mathcal{H}^0| \leq \Pi_{\mathcal{H}}(n-1)$. The set $\mathcal{H}^1 \setminus \mathcal{H}^0$ corresponds to restrictions of $\mathcal{H}$ to $\{x_1, \ldots, x_{n-1}\}$ where both labelings of x_{n-1} are achievable, which is restricted by $\mathrm{VC}(\mathcal{H}) - 1$: $|\mathcal{H}^1 \setminus \mathcal{H}^0| \leq \Pi_{\mathcal{H}'}(n-1)$ with $\mathrm{VC}(\mathcal{H}') \leq d - 1$. Combining: $\Pi_{\mathcal{H}}(n) \leq \Pi_{\mathcal{H}}(n-1) + \Pi_{\mathcal{H}'}(n-1) \leq \sum_{i=0}^{d} \binom{n-1}{i} + \sum_{i=0}^{d-1} \binom{n-1}{i} = \sum_{i=0}^{d} \binom{n}{i}$. $\square$

VC Generalization Bound

Let $\mathcal{H}$ have $\mathrm{VC}(\mathcal{H}) = d < \infty$. With probability $\geq 1 - \delta$:

$$\sup_{h \in \mathcal{H}} |R(h) - R_n(h)| \leq \sqrt{\frac{8d \log(2en/d) + 8 \log(4/\delta)}{n}}.$$

The proof uses Sauer's lemma to bound the growth function, then applies a union bound over the (polynomial number of) distinct labelings produced by $\mathcal{H}$, and finally applies Hoeffding within each labeling class.

Fundamental Theorem of Statistical Learning (Blumer et al., 1989; Ehrenfeucht et al., 1989)

For binary classification with 0-1 loss, the following are equivalent:

(i) $\mathcal{H}$ is agnostic PAC learnable.

(ii) $\mathcal{H}$ has the uniform convergence property.

(iii) $d = \mathrm{VC}(\mathcal{H}) < \infty$.

Moreover, the sample complexity $m_{\mathcal{H}}(\varepsilon, \delta)$ satisfies:

$$\Omega\left(\frac{d + \log(1/\delta)}{\varepsilon^2}\right) \leq m_{\mathcal{H}}(\varepsilon, \delta) \leq O\left(\frac{d \log(1/\varepsilon) + \log(1/\delta)}{\varepsilon^2}\right).$$

Practical consequence from hand notes. The VC bound is not data-dependent — it only uses the hypothesis class $\mathcal{H}$. This is both a strength (no assumptions on P) and a limitation (it may be very loose). In practice, we minimize $R_n(h) + \varepsilon_n(\mathcal{H}, \delta)$ over a *collection* of hypothesis classes $\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots$, selecting the best class via hyperparameter tuning (model selection). This is the **structural risk minimization** principle.

1.5. The Bias–Variance Decomposition

The Rademacher/VC framework controls the uniform deviation of empirical risk from population risk. A complementary — and perhaps more intuitive — decomposition splits the test error into *structural bias* and *random variance*.

Bias–Variance Decomposition (squared loss)

Let $f^*(x) = \mathbb{E}[Y|X = x]$ be the true regression function and $\hat{f}_n$ any estimator trained on data of size n. Then:

$$\mathbb{E}[(Y - \hat{f}_n(x))^2] = \underbrace{\left(f^*(x) - \mathbb{E}[\hat{f}_n(x)]\right)^2}_{\mathrm{Bias}^2(\hat{f}_n(x))} + \underbrace{\mathrm{Var}(\hat{f}_n(x))}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible noise}},$$

where $\sigma^2 = \mathrm{Var}(Y|X = x)$ and expectations are over $\mathcal{D}$ and Y.

Proof

Let $\bar{f}(x) = \mathbb{E}[\hat{f}_n(x)]$. Writing $Y = f^*(x) + \eta$ with $\mathbb{E}[\eta] = 0$, $\mathbb{E}[\eta^2] = \sigma^2$:

$$\begin{aligned}
\mathbb{E}[(Y - \hat{f}_n)^2] &= \mathbb{E}[(f^* + \eta - \hat{f}_n)^2] \\
&= \mathbb{E}[(f^* - \hat{f}_n)^2] + 2\mathbb{E}[(f^* - \hat{f}_n)\eta] + \sigma^2 \\
&= \mathbb{E}[(f^* - \hat{f}_n)^2] + \sigma^2 \quad (\hat{f}_n \perp\!\!\!\perp \eta \text{ given } x) \\
&= \mathbb{E}[(f^* - \bar{f} + \bar{f} - \hat{f}_n)^2] + \sigma^2 \\
&= (f^* - \bar{f})^2 + \mathbb{E}[(\hat{f}_n - \bar{f})^2] + 2(f^* - \bar{f})\underbrace{\mathbb{E}[\hat{f}_n - \bar{f}]}_{=0} + \sigma^2.
\end{aligned}$$

□

Example: Bias-Variance in Polynomial Regression

Figure 1.2 illustrates the tradeoff on a synthetic dataset where the true function is $f^*(x) = \sin(2\pi x)$ with Gaussian noise $\sigma = 0.3$. Degree-1 polynomials underfit (high bias, low variance). Degree-12 polynomials wildly overfit (low training error, high test error). An intermediate degree (around 4–6) achieves the best generalization.

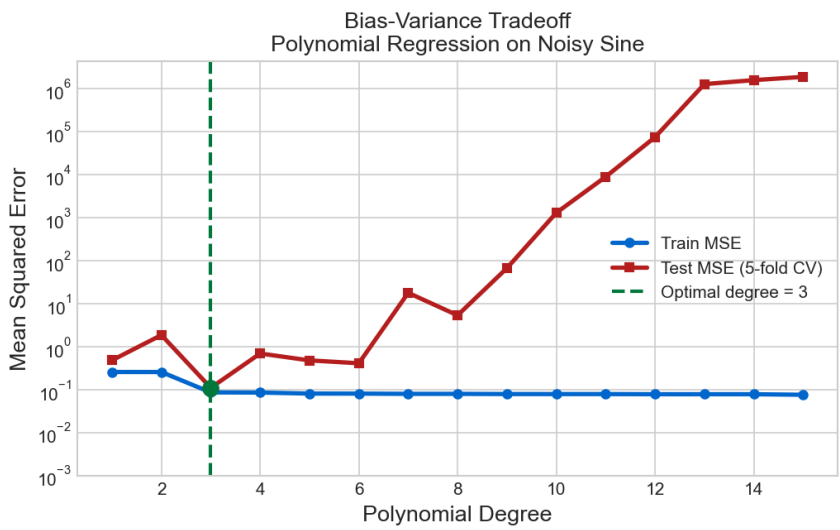


Figure 1.2: Train and test MSE as a function of polynomial degree on a synthetic dataset ($n = 200$, 5-fold CV). The gap between train and test error widens sharply for high degrees, illustrating overfitting.

Key insight. Bias is a structural property of the hypothesis class — it tells us how well the best function in $\mathcal{H}$ can approximate f^*. Variance is a statistical property — it tells us how much $\hat{f}_n$ fluctuates around its mean due to random sampling. Reducing bias typically requires enlarging $\mathcal{H}$ (more complex models), which increases variance. This is the *bias-variance tradeoff*, and it motivates regularization.

1.6. No Free Lunch and the Need for Priors

The No Free Lunch theorem (Wolpert, 1996) states that averaged over all possible data-generating distributions, every learning algorithm has the same expected error. This seems pessimistic: why does machine learning work in practice?

The answer is that real-world distributions are *not* arbitrary. Natural images, text, and financial data exhibit structure: smoothness, sparsity, low-dimensional manifolds, hierarchical features. Good machine learning methods succeed by implicitly or explicitly encoding the right inductive biases for the problems at hand. Regularization, architectural choices in neural networks, and kernel selection are all ways of specifying these biases.

Lecture 2

Regularisation

Historical Note: From OLS to Regularized Regression

Ordinary least squares dates to Gauss and Legendre (1805). Ridge regression was introduced by Hoerl and Kennard (1970) as a remedy for multicollinearity. The LASSO (Least Absolute Shrinkage and Selection Operator) was proposed by Tibshirani (1996), inspired partly by earlier work on basis pursuit in signal processing (Chen, Donoho, Saunders, 1998). The LARS algorithm for efficiently computing the LASSO path was developed by Efron, Hastie, Johnstone and Tibshirani (2004). Theoretical guarantees for LASSO under sparsity were established by Candès and Tao (2007) and Bickel, Ritov and Tsybakov (2009).

2.1. The Linear Model and Its Limits

Consider the linear model $Y = \boldsymbol{x}^\top \beta^* + \varepsilon$ with $\varepsilon \sim (0, \sigma^2)$ independent of $\boldsymbol{x} \in \mathbb{R}^p$. Let $\boldsymbol{X} \in \mathbb{R}^{n \times p}$ be the design matrix. The Ordinary Least Squares (OLS) estimator is:

$$\widehat{\beta}_{\text{OLS}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{Y},$$

which exists only when $p \leq n$ and $\boldsymbol{X}$ has full column rank.

When OLS fails.

1. $p > n$: OLS is undefined (the normal equations are underdetermined; infinitely many solutions).
2. Near multicollinearity ($p \leq n$ but columns of $\boldsymbol{X}$ nearly linearly dependent): $\boldsymbol{X}^\top \boldsymbol{X}$ is nearly singular, so $\left\|\widehat{\beta}_{\text{OLS}}\right\|$ is huge and $\widehat{\beta}_{\text{OLS}}$ has enormous variance.
3. Many irrelevant predictors: OLS uses all p variables even if most have $\beta_j^* = 0$, adding noise to estimates of non-zero coefficients.

The Gauss-Markov theorem says OLS is BLUE (Best Linear Unbiased Estimator) — but unbiasedness is not always desirable. By tolerating a small bias, we can dramatically reduce variance and improve mean squared error.

2.1.1. The Parametric Rate

OLS Parametric Rate

Under $Y = \boldsymbol{X}\beta^* + \varepsilon$, $\varepsilon \sim (0, \sigma^2 \boldsymbol{I})$:

$$\widehat{\beta}_{\text{OLS}} - \beta^* = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \varepsilon.$$

Hence $\mathbb{E}[\left\|\widehat{\beta} - \beta^*\right\|_2^2] = \sigma^2 \operatorname{tr}((\boldsymbol{X}^\top \boldsymbol{X})^{-1}) = O(p/n)$, giving $\left\|\widehat{\beta} - \beta^*\right\|_2 = O_{\mathbb{P}}(\sqrt{p/n})$.

Proof

$\widehat{\beta} - \beta^* = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \varepsilon$. Then:

$$\mathbb{E}[\left\|\widehat{\beta} - \beta^*\right\|^2] = \mathbb{E}[\varepsilon^\top \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-2} \boldsymbol{X}^\top \varepsilon] = \sigma^2 \operatorname{tr}((\boldsymbol{X}^\top \boldsymbol{X})^{-1}).$$

For isotropic design ($\boldsymbol{X}^\top \boldsymbol{X}/n \approx \boldsymbol{I}_p$): $\operatorname{tr}((\boldsymbol{X}^\top \boldsymbol{X})^{-1}) \approx p/n$. □

When $p > n$, OLS is undefined ($\boldsymbol{X}^\top \boldsymbol{X}$ is singular). Even for $p = n/2$, the rate $\sqrt{p/n} = 1/\sqrt{2}$ is too slow for high-quality estimation. We need *structural assumptions* (sparsity, smoothness) and corresponding regularization to do better.

2.1.2. Minimax Lower Bounds

A natural question: for a given class of functions (e.g., s-sparse vectors), what is the *best possible* rate that *any* estimator can achieve? This is the minimax rate.

Minimax Rate for Sparse Linear Regression

Over the class $\Sigma(s, p) = \{\beta \in \mathbb{R}^p : \|\beta\|_0 \leq s\}$ with isotropic Gaussian design:

$$\inf_{\hat{\beta}} \sup_{\beta^* \in \Sigma(s,p)} \mathbb{E}[\left\|\hat{\beta} - \beta^*\right\|^2] \asymp \frac{s \log(p/s)}{n}.$$

Lower bound proof sketch (Varshamov–Gilbert + Fano)

Step 1. **Discretize**: construct a finite packing $\{S_0, \ldots, S_M\} \subset \Sigma(s, p)$ with $\|S_j - S_k\|^2 \geq c$ for all $j \neq k$ using the Varshamov–Gilbert bound, yielding $M \geq 2^{s \log(p/s)/8}$.

Step 2. **Feature space reduction**: embed the s-sparse problem in $m = \binom{p}{s}$ coordinate-aligned hypercubes of side length h, giving $m \lesssim (p/s)^s$ local alternatives.

Step 3. **KL control**: for Gaussian noise, $\mathrm{KL}(P_j \| P_k) \leq n \|S_j - S_k\|^2 / (2\sigma^2)$.

Step 4. **Fano's inequality**: if M hypotheses have pairwise KL bounded by β, any test has error $\geq 1 - (\beta + \log 2)/\log M$. Choosing the separation to balance gives the $s \log(p/s)/n$ rate.

This shows that the LASSO rate $\sqrt{s \log p/n}$ is nearly optimal (up to the $\log(p/s)$ vs. $\log p$ distinction). □

Remark from hand notes: Random Forests beat the minimax rate

The nonparametric minimax rate for a Lipschitz function in $\mathbb{R}^d$ is $n^{-2/(d+2)}$. When the function has additional structure — sparsity in relevant variables, low intrinsic dimensionality, or lies on a manifold — random forests can achieve rates that depend on the effective (lower) dimension rather than the ambient d. This "implicit manifold" property gives random forests an advantage over classical nonparametric estimators that are cursed by the ambient dimension.

2.2. Ridge Regression

Definition: Ridge Regression (Tikhonov Regularization)

The ridge estimator adds an ℓ_2 penalty to the OLS objective:

$$\widehat{\beta}_{\text{ridge}} = \underset{\beta \in \mathbb{R}^p}{\arg\min} \left\{ \|\boldsymbol{Y} - \boldsymbol{X}\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}, \quad \lambda > 0.$$

This has the closed-form solution:

$$\widehat{\beta}_{\text{ridge}} = (\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I})^{-1} \boldsymbol{X}^\top \boldsymbol{Y}.$$

Derivation of Ridge closed form

The objective $\mathcal{L}(\beta) = \|\boldsymbol{Y} - \boldsymbol{X}\beta\|^2 + \lambda \|\beta\|^2$ is strictly convex. Setting the gradient to zero:

$$\nabla_\beta \mathcal{L} = -2\boldsymbol{X}^\top(\boldsymbol{Y} - \boldsymbol{X}\beta) + 2\lambda\beta = 0 \implies (\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I})\beta = \boldsymbol{X}^\top \boldsymbol{Y}.$$

Since $\boldsymbol{X}^\top \boldsymbol{X}$ is positive semi-definite and $\lambda > 0$, the matrix $\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}$ is positive definite, hence invertible, giving the formula. $\square$

2.2.1. Singular Value Analysis of Ridge

Let $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top$ be the SVD with singular values $d_1 \geq \cdots \geq d_p \geq 0$. Then:

$$\widehat{\beta}_{\text{OLS}} = \sum_{j=1}^{p} \frac{d_j}{d_j^2} \boldsymbol{v}_j \boldsymbol{u}_j^\top \boldsymbol{Y} = \sum_{j=1}^{p} \frac{1}{d_j} \boldsymbol{v}_j \boldsymbol{u}_j^\top \boldsymbol{Y},$$

$$\widehat{\beta}_{\text{ridge}} = \sum_{j=1}^{p} \frac{d_j}{d_j^2 + \lambda} \boldsymbol{v}_j \boldsymbol{u}_j^\top \boldsymbol{Y}.$$

Ridge multiplies the contribution of each singular direction $\boldsymbol{v}_j$ by the **shrinkage factor** $d_j^2/(d_j^2 + \lambda)$. Directions with small singular value (near-multicollinear directions) are shrunk strongly toward zero; dominant directions are barely affected.

2.2.2. Bias-Variance Analysis of Ridge

Bias and Variance of Ridge

$$\mathrm{Bias}(\widehat{\beta}_{\text{ridge}}) = -\lambda(\boldsymbol{X}^\top\boldsymbol{X}+\lambda\boldsymbol{I})^{-1}\beta^*, \quad \mathrm{Var}(\widehat{\beta}_{\text{ridge}}) = \sigma^2(\boldsymbol{X}^\top\boldsymbol{X}+\lambda\boldsymbol{I})^{-1}\boldsymbol{X}^\top\boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X}+\lambda\boldsymbol{I})^{-1}.$$

Proof

Let $\boldsymbol{A} = (\boldsymbol{X}^\top\boldsymbol{X} + \lambda\boldsymbol{I})^{-1}\boldsymbol{X}^\top$. Then $\widehat{\beta}_{\text{ridge}} = \boldsymbol{A}\boldsymbol{Y} = \boldsymbol{A}(\boldsymbol{X}\beta^* + \varepsilon)$. So:

$$\mathbb{E}[\widehat{\beta}_{\text{ridge}}] = \boldsymbol{A}\boldsymbol{X}\beta^* = (\boldsymbol{X}^\top\boldsymbol{X} + \lambda\boldsymbol{I})^{-1}\boldsymbol{X}^\top\boldsymbol{X}\beta^* = \beta^* - \lambda(\boldsymbol{X}^\top\boldsymbol{X} + \lambda\boldsymbol{I})^{-1}\beta^*.$$

The variance follows from $\mathrm{Var}(\boldsymbol{A}\varepsilon) = \sigma^2\boldsymbol{A}\boldsymbol{A}^\top$. □

There exists a $\lambda^* > 0$ for which the ridge MSE is strictly smaller than OLS MSE. As $\lambda \to 0$, ridge approaches OLS (unbiased, high variance); as $\lambda \to \infty$, ridge shrinks to zero (highly biased, zero variance). The optimal λ balances these via cross-validation.

2.2.3. Bayesian Interpretation

Ridge is equivalent to Bayesian linear regression with a Gaussian prior $\beta \sim \mathcal{N}(0, \tau^2\boldsymbol{I})$: the ridge estimator with $\lambda = \sigma^2/\tau^2$ is the posterior mean. This connection makes ridge regularization interpretable as incorporating prior belief that coefficients are small.

2.2.4. Oracle Inequality for Ridge

A key strength of ridge: the excess risk bound is *dimension-independent*.

Oracle Inequality for Ridge (Shalev-Shwartz & Ben-David, Cor. 13.8)

Assume the loss ℓ is convex and ρ-Lipschitz, and $\|x_i\| \le 1$. The ridge objective $\mathcal{L}(\beta) = R_n(\beta) + \lambda\|\beta\|^2$ is 2λ-strongly convex. Then:

$$\mathbb{E}[R(\widehat{\beta}_{\text{ridge}})] \le R(\beta^*) + \lambda\|\beta^*\|^2 + \frac{\rho^2}{2\lambda\sqrt{n}}.$$

Proof

Since ERM with 2λ-strongly convex regularization achieves uniform deviation $\mathbb{E}[R(\widehat{\beta}) - R_n(\widehat{\beta})] \le \rho^2/(2\lambda\sqrt{n})$ (from covering number arguments for norm-bounded classes; the key is that $\left\|\widehat{\beta}_{\text{ridge}}\right\| \le \|\beta^*\|$ by optimality). By the ERM guarantee:

$$R_n(\widehat{\beta}_{\text{ridge}}) + \lambda\left\|\widehat{\beta}_{\text{ridge}}\right\|^2 \le R_n(\beta^*) + \lambda\|\beta^*\|^2.$$

Adding the generalization bound: $\mathbb{E}[R(\widehat{\beta}_{\text{ridge}})] \le R(\beta^*) + \lambda\|\beta^*\|^2 + \rho^2/(2\lambda\sqrt{n})$. □

Optimizing λ. Setting $\lambda^* = \rho/(\|\beta^*\| \cdot n^{1/4})$:

$$\mathbb{E}[R(\widehat{\beta}_{\text{ridge}})] - R(\beta^*) \le \frac{\rho\|\beta^*\|}{\sqrt[4]{n}} = O\left(\frac{\rho\|\beta^*\|}{n^{1/4}}\right).$$

This is **independent of the dimension** p — only $\|\beta^*\|$ matters. Ridge with optimal λ

always improves upon OLS.

2.2.5. Support Recovery Conditions for LASSO

The LASSO rate theorem requires the RE condition. Here we place it within the broader landscape of conditions:

Hierarchy of Conditions for LASSO

(i) **RIP** (Restricted Isometry Property, Candès & Tao): $(1-\delta)\|\beta\|^2 \le \frac{1}{n}\|\boldsymbol{X}\beta\|^2 \le (1+\delta)\|\beta\|^2$ for all s-sparse β. Every s-column submatrix of $\boldsymbol{X}$ behaves like an orthogonal matrix. Strongest but hardest to verify.

(ii) **RE** (Restricted Eigenvalue, Bickel-Ritov-Tsybakov): $\frac{1}{n}\|\boldsymbol{X}v\|^2 \ge \kappa\|v_S\|^2$ only for v in the **cone** $\mathcal{C}(S, c_0) = \{v : \|v_{S^c}\|_1 \le c_0\|v_S\|_1\}$. Weaker than RIP; the LASSO naturally forces the estimation error $\Delta = \widehat{\beta} - \beta^*$ into this cone.

(iii) **Cone condition**: since β^* is s-sparse ($\beta^*_{S^c} = 0$), the basic inequality of LASSO forces $\|\Delta_{S^c}\|_1 \le 3\|\Delta_S\|_1$, so $\Delta \in \mathcal{C}(S, 3)$. The ℓ_1 penalty geometrically forces the error into a narrow cone around the support.

(iv) **IRC** (Irrepresentable Condition): $\left\|\boldsymbol{X}_{S^c}^\top \boldsymbol{X}_S(\boldsymbol{X}_S^\top \boldsymbol{X}_S)^{-1}\right\|_\infty < 1$. Required for *exact support recovery* ($\mathrm{supp}(\widehat{\beta}) = S$), not just estimation. The extra variables outside S must not be too highly correlated with the active variables.

Hierarchy: RIP $\Rightarrow$ RE $\Rightarrow$ prediction consistency. IRC is orthogonal (needed for model selection, not rate).

2.3. The LASSO

Ridge never sets coefficients to exactly zero, making the model hard to interpret in high dimensions. The LASSO addresses this with an ℓ_1 penalty.

Definition: LASSO (Tibshirani, 1996)

$$\widehat{\beta}_{\text{lasso}} = \arg\min_{\beta}\left\{\frac{1}{2n}\|\boldsymbol{Y} - \boldsymbol{X}\beta\|_2^2 + \lambda\|\beta\|_1\right\}.$$

Unlike ridge, this convex but non-smooth optimization problem has no closed form in general. However, for orthogonal designs ($\boldsymbol{X}^\top\boldsymbol{X} = n\boldsymbol{I}$), the solution is explicit.

LASSO under Orthogonal Design: Soft Thresholding

If $\boldsymbol{X}^\top\boldsymbol{X} = n\boldsymbol{I}$, then:

$$\widehat{\beta}_j^{\text{lasso}} = S_\lambda(\widehat{\beta}_j^{\text{OLS}}),$$

where the **soft-thresholding operator** is:

$$S_\lambda(z) = \mathrm{sign}(z)\cdot(|z| - \lambda)_+ = \begin{cases} z - \lambda & z > \lambda \\ 0 & |z| \le \lambda \\ z + \lambda & z < -\lambda. \end{cases}$$

Proof

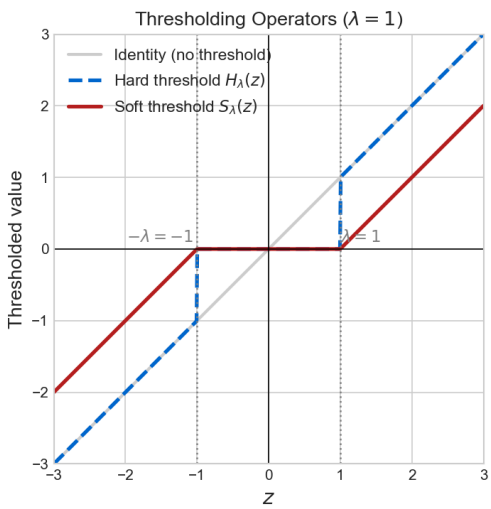
Under orthogonal design, the LASSO objective decouples coordinate-wise:

$$\frac{1}{2n}\|\boldsymbol{Y}-\boldsymbol{X}\beta\|^2+\lambda\|\beta\|_1=\sum_{j=1}^{p}\left[\frac{1}{2}(\hat{\beta}_j^{\mathrm{OLS}}-\beta_j)^2+\lambda|\beta_j|\right]+\text{const.}$$

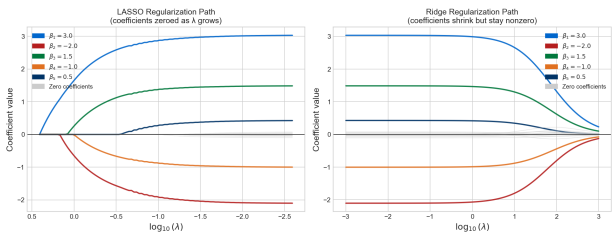
Each term $g_j(\beta_j)=\frac{1}{2}(\hat{\beta}_j-\beta_j)^2+\lambda|\beta_j|$ is minimized separately. The subdifferential condition $0\in\partial g_j$ gives:

- If $\hat{\beta}_j>\lambda$: $\beta_j=\hat{\beta}_j-\lambda$ (shift down by λ).
- If $|\hat{\beta}_j|\le\lambda$: $\beta_j=0$ (zero is in the subdifferential since $|\hat{\beta}_j-0|=|\hat{\beta}_j|\le\lambda$).
- If $\hat{\beta}_j<-\lambda$: $\beta_j=\hat{\beta}_j+\lambda$ (shift up by λ).

□



(a) Soft thresholding operator.



(b) Regularization paths: LASSO (left) vs. Ridge (right).

Figure 2.1: (a) The soft-thresholding operator $S_\lambda(z)$ sets small inputs to exactly zero, unlike hard thresholding which creates discontinuities. (b) As λ increases, LASSO coefficients hit zero at finite λ (sparsity), while Ridge coefficients only approach zero asymptotically.

Why does ℓ_1 create zeros but ℓ_2 does not? The ℓ_1 ball $\{\beta:\|\beta\|_1\le t\}$ has corners at the coordinate axes. The level sets of the squared loss are ellipsoids. When these ellipsoids are expanded outward from $\hat{\beta}^{\mathrm{OLS}}$, they generically hit a corner of the ℓ_1 ball first — a corner where some coordinates are exactly zero. The ℓ_2 ball is smooth everywhere, so the tangency is never at a sparse solution.

2.3.1. Coordinate Descent for LASSO

Definition: Coordinate Descent Algorithm

Initialize $\beta^{(0)}=\mathbf{0}$. Repeat until convergence:

1. For $j=1,\dots,p$:
2. Compute the partial residual: $r_j=\boldsymbol{Y}-\sum_{k\neq j}x_k\beta_k$.
3. Update: $\beta_j\leftarrow S_{n\lambda}(x_j^\top r_j/\|x_j\|^2)/(\|x_j\|^2/n)$.

Each coordinate update has a closed form (soft-thresholding applied to the partial regression coefficient), so each cycle costs $O(np)$. Convergence to the global optimum is guaranteed because the LASSO objective is convex and coordinate-wise updates are exact. Pathological cycling cannot occur.

2.3.2. Statistical Theory: Sparsity and Rates

The LASSO's statistical power comes from sparsity: if β^* has only $s \ll p$ nonzero entries, the LASSO recovers it at the optimal rate.

Definition: Restricted Eigenvalue Condition

The design matrix $\boldsymbol{X}$ satisfies $\mathrm{RE}(s, c_0)$ if there exists $\kappa > 0$ such that for all $\beta \neq 0$ with $\|\beta_{S^c}\|_1 \leq c_0 \|\beta_S\|_1$ (where $S = \mathrm{supp}(\beta^*)$, $|S| = s$):

$$\frac{1}{n} \|\boldsymbol{X}\beta\|_2^2 \geq \kappa \|\beta_S\|_2^2 .$$

LASSO Rate Theorem (Bickel, Ritov, Tsybakov, 2009)

Suppose $\boldsymbol{X}$ satisfies $\mathrm{RE}(s, 3)$ with constant κ, $\varepsilon \sim \mathcal{N}(0, \sigma^2 \boldsymbol{I})$, and $\lambda = c\sigma\sqrt{\log p/n}$ for a sufficiently large constant c. Then with probability $\geq 1 - 2/p$:

$$\left\|\widehat{\beta}_{\text{lasso}} - \beta^*\right\|_2 \leq \frac{4c\sigma}{\kappa}\sqrt{\frac{s \log p}{n}}.$$

Proof sketch

Let $\Delta = \widehat{\beta} - \beta^*$. The **basic inequality** of LASSO: since $\widehat{\beta}$ minimizes the objective and β^* is feasible,

$$\frac{1}{2n}\left\|\boldsymbol{X}\widehat{\beta} - \boldsymbol{Y}\right\|^2 + \lambda\left\|\widehat{\beta}\right\|_1 \leq \frac{1}{2n}\|\boldsymbol{X}\beta^* - \boldsymbol{Y}\|^2 + \lambda\|\beta^*\|_1 .$$

Expanding (with $\boldsymbol{Y} = \boldsymbol{X}\beta^* + \varepsilon$) and using $\left\|\widehat{\beta}\right\|_1 \geq \|\beta^*_S\|_1 - \|\Delta_S\|_1 + \|\Delta_{S^c}\|_1$ gives:

$$\frac{1}{2n}\|\boldsymbol{X}\Delta\|^2 \leq \frac{1}{n}\varepsilon^\top \boldsymbol{X}\Delta + \lambda(\|\Delta_S\|_1 - \|\Delta_{S^c}\|_1).$$

Control the noise term. With probability $\geq 1 - 2/p$, $\left\|\boldsymbol{X}^\top\varepsilon/n\right\|_\infty \leq c\sigma\sqrt{\log p/n}$ (union bound over columns + Gaussian tail). Choosing $\lambda = c\sigma\sqrt{\log p/n}$ and using Cauchy-Schwarz: $\frac{1}{n}\varepsilon^\top \boldsymbol{X}\Delta \leq \lambda\|\Delta\|_1$.
Cone condition. The basic inequality then forces $\|\Delta_{S^c}\|_1 \leq 3\|\Delta_S\|_1$, so Δ lies in the restricted set where $\mathrm{RE}(s, 3)$ applies.
Apply RE. $\frac{\kappa}{2}\|\Delta_S\|^2 \leq \frac{1}{2n}\|\boldsymbol{X}\Delta\|^2 \leq 2\lambda\|\Delta_S\|_1 \leq 2\lambda\sqrt{s}\|\Delta_S\|_2$. Dividing: $\|\Delta\|_2 \leq \|\Delta_S\|_2 + \|\Delta_{S^c}\|_2 \lesssim \|\Delta_S\|_2 \leq 4\lambda\sqrt{s}/\kappa$. □

The Rate $\sqrt{s \log p/n}$ and Its Importance

The LASSO achieves estimation error $O(\sqrt{s \log p/n})$. This is *nearly* minimax optimal for sparse estimation (the minimax lower bound is $\Omega(\sqrt{s \log(p/s)/n})$). The $\log p$

factor is the unavoidable price of model selection over p variables. We will return to this rate in Lecture 8, where it determines whether DML achieves valid $\sqrt{n}$-inference.

Example: Application: Genetic Association Studies

In genome-wide association studies (GWAS), researchers test whether individual genetic variants (SNPs) predict a phenotype (e.g., height, disease risk). A typical study has $n \approx 10{,}000$ individuals and $p \approx 500{,}000$ SNPs. Only a handful of SNPs ($s \ll p$) are truly associated. LASSO is widely used to identify these sparse associations, with λ chosen by cross-validation. The rate $\sqrt{s \log p/n} \approx \sqrt{s \cdot 13/10{,}000}$ shows that even $n = 10{,}000$ can recover a moderate number of true signals.

2.4. Model Selection: Cross-Validation and the Regularisation Path

For both Ridge and LASSO, λ is a hyperparameter that must be chosen. Cross-validation is the standard method.

K-fold cross-validation. Split the data into K folds $\mathcal{D}_1, \ldots, \mathcal{D}_K$. For each λ:

1. For $k = 1, \ldots, K$: train on $\mathcal{D} \setminus \mathcal{D}_k$, evaluate on $\mathcal{D}_k$.
2. Average the validation errors: $\mathrm{CV}(\lambda) = \frac{1}{K} \sum_{k=1}^{K} \hat{R}_{\mathcal{D}_k}(\widehat{f}^{\lambda})$.

Choose $\hat{\lambda} = \arg\min_{\lambda} \mathrm{CV}(\lambda)$.

The **regularisation path** is the curve $\lambda \mapsto \widehat{\beta}(\lambda)$. For LASSO, as λ decreases from $\lambda_{\max} = \left\| \boldsymbol{X}^\top \boldsymbol{Y}/n \right\|_\infty$ (all coefficients zero) to zero (OLS), coefficients enter the model one by one. The path is piecewise linear and can be traced exactly via the LARS algorithm. The LARS algorithm adds the variable most correlated with the current residual, traces the coefficient along a direction equiangular between the active set, and adds the next variable when a tie occurs.

Lecture 3

Decision Trees and Random Forests

Historical Note: From Single Trees to Forests

Decision trees have roots in the social sciences (Morgan & Sonquist, 1963, AID algorithm) and statistics (Breiman, Friedman, Olshen & Stone, 1984, CART). For decades, they were considered the state of the art for interpretable models. The idea of combining trees emerged with bagging (Breiman, 1996) and was perfected in random forests (Breiman, 2001), one of the most widely used methods in practice. The first rigorous consistency proof for CART-based random forests was given by Scornet, Biau and Vert (2015), over a decade after the method became widely used.

3.1. Decision Trees

Definition: Regression Tree

A regression tree partitions $\mathcal{X} \subseteq \mathbb{R}^p$ into M rectangular (axis-aligned) regions $R_1, \ldots, R_M$ and predicts a constant in each:

$$\hat{f}(\boldsymbol{x}) = \sum_{m=1}^{M} \hat{c}_m \, \mathbb{1}[\boldsymbol{x} \in R_m], \qquad \hat{c}_m = \frac{1}{|R_m|} \sum_{i:\boldsymbol{x}_i \in R_m} y_i.$$

The prediction in each leaf is the regional average of training labels, which is the Bayes-optimal constant predictor for squared loss within that region.

3.1.1. Why Trees?

Trees have several attractive properties:

1. **Interpretability**: any tree is equivalent to a flowchart of yes/no questions on individual features. A depth-5 tree requires at most 5 questions to classify any input.
2. **Robustness to feature scaling**: splitting criteria depend only on rank ordering within each feature, not absolute values. Features need not be standardized.
3. **Handles categorical variables and missing data**: CART handles both natively via surrogate splits.

4. **Captures non-linear interactions**: a depth-d tree can represent 2^d interaction patterns.
5. **Nonparametric**: no distributional assumptions on the data.

The main weakness: high variance. Small perturbations of the training data yield completely different trees, because the greedy splitting is unstable.

3.1.2. The CART Algorithm

Definition: CART Splitting Criterion (Regression)

At a node with data $\{(\boldsymbol{x}_i, y_i) : i \in I\}$, find the best feature j and cutpoint s:

$$(j^*, s^*) = \underset{j \in [p],\, s \in \mathbb{R}}{\arg\min} \left[\sum_{i \in I : x_{ij} \le s} (y_i - \hat{c}_L)^2 + \sum_{i \in I : x_{ij} > s} (y_i - \hat{c}_R)^2 \right],$$

where $\hat{c}_L$ and $\hat{c}_R$ are the means of y_i in the left and right child nodes.

At each node, we try all p features and all $O(n)$ candidate cutpoints (unique values of each feature). Finding the optimal split takes $O(np)$ time, and growing a full tree to n leaves takes $O(n^2 p)$ time in the worst case (though $O(np \log n)$ with sorted data).

Classification trees replace the residual sum of squares with an impurity measure. Three common choices at a node m with class proportions $\hat{p}_{mk}$:

- **Gini index**: $G(m) = \sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk})$. Zero for pure nodes.
- **Cross-entropy**: $H(m) = -\sum_k \hat{p}_{mk} \log \hat{p}_{mk}$. More sensitive to imbalanced splits.
- **Misclassification rate**: $1 - \max_k \hat{p}_{mk}$. Simple but not differentiable.

Gini and cross-entropy both equal zero for pure nodes ($\hat{p}_{mk} = 1$ for some k) and are maximized at uniform distributions. They are preferred over misclassification rate because they are more sensitive to improving purity.

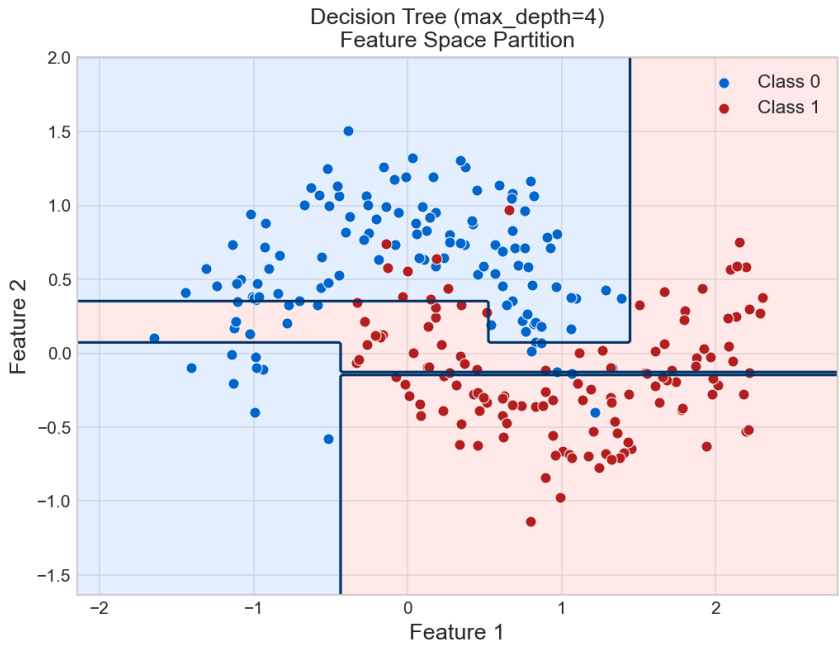


Figure 3.1: Feature space partition induced by a depth-4 decision tree on the two-moons dataset. Each region is labeled by its majority class. Axis-aligned boundaries are a limitation — the true boundary is curved.

3.1.3. The Bias-Variance Tradeoff in Trees

Bias-Variance in Decision Trees

For a tree of depth d:

$$\mathbb{E}[(y - \hat{f}(\boldsymbol{x}))^2] = \text{Bias}^2[\hat{f}(\boldsymbol{x})] + \text{Var}(\hat{f}(\boldsymbol{x})) + \sigma^2.$$

- Deep trees (d large): low bias (fine partitions approximate f^* well), high variance (regions contain few points, means are noisy).
- Shallow trees (d small): high bias (coarse partitions cannot approximate f^*), low variance (regions contain many points, means are stable).

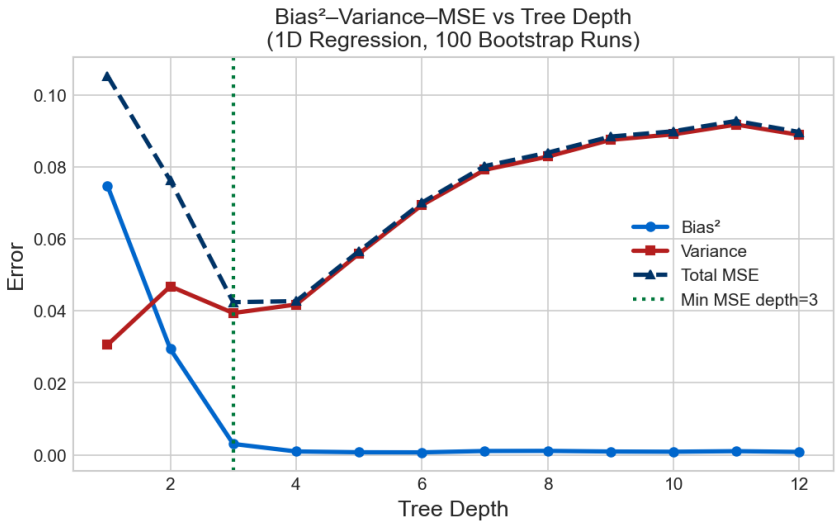


Figure 3.2: Squared bias, variance, and total test MSE vs. tree depth on a 1D regression problem. Variance grows monotonically with depth; bias decreases. The optimal depth balances these two.

3.1.4. Cost-Complexity Pruning

A fully grown tree overfits. **Cost-complexity pruning** (weakest-link pruning, Breiman 1984) finds the optimal subtree for any $\alpha \geq 0$ by penalizing tree size:

$$C_\alpha(T) = \underbrace{\sum_{m=1}^{|T|} \sum_{\boldsymbol{x}_i \in R_m} (y_i - \hat{c}_m)^2}_{\text{training RSS}} + \alpha |T|.$$

The algorithm works by iteratively collapsing the internal node whose removal increases $C_0(T)$ least, producing a nested sequence $T_{\max} \supset T_1 \supset \cdots \supset T_0$ (root only). The optimal α — and hence the optimal subtree — is selected by cross-validation.

The complexity penalty α plays the same role as λ in LASSO: larger values produce simpler, more regularized models.

3.1.5. Minimax Rate for Trees

For a Lipschitz function f^* on $\mathbb{R}^d$, any tree estimator achieves at best:

$$\mathbb{E}\left[\left\|\hat{f}_n - f^*\right\|^2\right] = O\left(n^{-2/(d+2)}\right).$$

This is the same rate as a k-nearest neighbor estimator and suffers the curse of dimensionality: in $d = 10$ dimensions, going from $n = 1000$ to $n = 1{,}000{,}000$ reduces the error by only a factor of ≈ 4.

3.2. Ensemble Methods

3.2.1. The Variance Reduction Argument

The key insight that motivates ensembles is the following calculation. Suppose $\hat{f}_1, \ldots, \hat{f}_B$ are estimators that are identically distributed (same marginal distribution), with pairwise

correlation $\rho = \mathrm{Corr}(\hat{f}_b(\boldsymbol{x}), \hat{f}_{b'}(\boldsymbol{x}))$ and individual variance σ^2. The averaged estimator $\bar{f} = \frac{1}{B}\sum_b \hat{f}_b$ has:

$$\mathrm{Var}(\bar{f}(\boldsymbol{x})) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

Proof

$$\begin{aligned}\mathrm{Var}\left(\frac{1}{B}\sum_b \hat{f}_b\right) &= \frac{1}{B^2}\sum_b \mathrm{Var}(\hat{f}_b) + \frac{1}{B^2}\sum_{b\neq b'} \mathrm{Cov}(\hat{f}_b, \hat{f}_{b'}) \\ &= \frac{\sigma^2}{B} + \frac{B(B-1)}{B^2}\rho\sigma^2 = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.\end{aligned}$$

□

As $B \to \infty$, the variance floor is $\rho\sigma^2$ — the *irreducible* correlation between any two trees. Reducing ρ reduces the floor. This is the motivation for both bagging and random forests.

3.2.2. Bagging

Bootstrap aggregation (bagging, Breiman 1996) generates diverse trees by training each on an independent bootstrap sample. A bootstrap sample $\mathcal{D}_b^*$ is obtained by sampling n observations from $\mathcal{D}$ uniformly with replacement. Predictions are then averaged.

Why does bootstrap help? Each tree sees a slightly different dataset, so the trees differ in their high-variance fitting of noise. The differences are uncorrelated enough to reduce variance upon averaging, while the bias of each tree is approximately the same as training on the full dataset (bootstrap samples have similar statistics to the original).

3.2.3. Out-of-Bag Error Estimation

Each bootstrap sample omits approximately $1/e \approx 36.8\%$ of observations. The *out-of-bag* (OOB) estimator evaluates each observation only on trees that did not include it in training:

$$\widehat{\mathrm{Err}}^{\mathrm{OOB}} = \frac{1}{n}\sum_{i=1}^{n} L\left(y_i, \frac{1}{|\mathcal{B}_i|}\sum_{b\in\mathcal{B}_i} T_b(\boldsymbol{x}_i)\right),$$

where $\mathcal{B}_i = \{b : i \notin \mathcal{D}_b^*\}$. This is an approximately unbiased estimate of the generalization error, with no need for a separate validation set.

3.3. Random Forests

Definition: Breiman's Random Forest Algorithm

For $b = 1, \ldots, B$:

1. Draw a bootstrap sample $\mathcal{D}_b^* \sim \mathcal{D}$ (with replacement, size n).
2. Grow a regression tree T_b on $\mathcal{D}_b^*$ using the modified splitting rule: at each node, select m features uniformly at random from the p available, and split optimally among only those m features.
3. Grow the tree to maximum depth (or to a minimum leaf size).

Aggregate: $\hat{f}^{\mathrm{RF}}(\boldsymbol{x}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\boldsymbol{x})$.

The random feature subset m (typically $m = \lfloor\sqrt{p}\rfloor$ for classification, $m = \lfloor p/3\rfloor$ for regression) is the key innovation. It forces trees to split on different features, drastically reducing inter-tree correlation ρ below what bagging achieves. The bias increases slightly (we may miss the globally best split), but the variance reduction dominates.

3.3.1. Variable Importance

A practically important by-product of random forests is a measure of variable importance, which cannot be computed from a single tree. Two measures:

Mean Decrease in Impurity (MDI). For each feature j, sum the improvement in splitting criterion over all nodes in all trees that split on feature j:

$$\mathrm{VI}_j^{\mathrm{MDI}} = \frac{1}{B}\sum_{b=1}^{B}\sum_{t:j(t)=j} |\tilde{R}_t| \cdot \Delta_t,$$

where Δ_t is the impurity decrease at node t and $|\tilde{R}_t|$ is the proportion of training samples reaching t. Simple to compute, but biased toward high-cardinality features.

Mean Decrease in Accuracy (MDA / permutation importance). For each tree b, permute the values of feature j in the OOB samples and measure the increase in OOB error:

$$\mathrm{VI}_j^{\mathrm{MDA}} = \frac{1}{B}\sum_{b=1}^{B}(\widehat{\mathrm{Err}}_j^{(b)} - \widehat{\mathrm{Err}}_{\mathrm{OOB}}^{(b)}),$$

where $\widehat{\mathrm{Err}}_j^{(b)}$ is the OOB error after shuffling column j. This is model-agnostic and unbiased.

SHAP values (Lundberg & Lee, 2017). A game-theoretic approach based on Shapley values from cooperative game theory. For each prediction $\hat{f}(x)$, the SHAP value $\phi_j(x)$ measures the marginal contribution of feature j to the prediction, averaged over all possible orderings of feature inclusion. For tree-based models, SHAP values can be computed in $O(TL^2)$ per prediction (where T is the number of trees, L is the maximum depth), making them practical for random forests. Unlike MDI, SHAP values are *local* (per-prediction) and *consistent* (if a model changes so that a feature's contribution increases, the SHAP value never decreases).

3.3.2. Causal Forests

Wager and Athey (2018) extended random forests to estimate heterogeneous treatment effects $\tau(\boldsymbol{x}) = \mathbb{E}[Y(1) - Y(0)|\boldsymbol{X} = \boldsymbol{x}]$ (the CATE). The key modifications:

1. **Honest trees**: use one half of the data for splitting, another for estimating leaf means. This eliminates the overfitting-induced bias in leaf estimates.

2. **CATE-targeted splits**: split to minimize variance of the CATE estimate, not outcome variance.

3. **Asymptotic normality**: under honesty and regularity, each leaf estimate is asymptotically normal, enabling pointwise confidence intervals for $\tau(\boldsymbol{x})$.

3.3.3. Consistency Results

Consistency of Random Forests (Scornet, Biau & Vert, 2015)

For CART-based random forests on additive models $f^*(\boldsymbol{x}) = \sum_{j=1}^{p} f_j(x_j)$ with independent features:

$$\mathbb{E}[(\hat{f}_n(\boldsymbol{x}) - f^*(\boldsymbol{x}))^2] \xrightarrow{n\to\infty} 0.$$

Proof sketch

Under the additive model, each coordinate is effectively a univariate problem. The key steps are: (1) show that the random forest estimator can be decomposed into a sum of univariate kernel estimators; (2) each univariate kernel has bandwidth shrinking to zero as $n \to \infty$ and $k_n \to \infty$ (minimum leaf size); (3) apply standard kernel regression consistency theory. The full proof controls the error from the bootstrap (showing bootstrap approximates in-sample training) and from the random feature selection (showing each coordinate is eventually selected). □

For Hölder-smooth f^* with smoothness parameter β in $\mathbb{R}^d$, the best known adaptable rate for random forests is $\tilde{O}(n^{-\beta/(2\beta+d)})$. This rate will reappear in Lecture 8, where it determines when DML can use random forests for valid causal inference.

3.4. Benign Overfitting and Double Descent

Historical Note: The Double Descent Discovery

Classical statistics and the bias-variance tradeoff predict that test error should increase monotonically with model complexity beyond a certain point. In 2018–2019, Belkin, Hsu, Ma and Mandal observed empirically — and Bartlett, Montanari and Rakhlin later proved rigorously for linear models — that test error in fact decreases again after the interpolation threshold $p = n$. This phenomenon was called *double descent* and forced a fundamental rethinking of the bias-variance tradeoff in modern deep learning.

Double Descent (Belkin et al., 2019; Bartlett et al., 2020)

Consider linear regression $Y = \boldsymbol{x}^\top \beta^* + \varepsilon$ with $\boldsymbol{x} \sim \mathcal{N}(0, \boldsymbol{\Sigma})$. The minimum-norm interpolating estimator $\widehat{\beta} = \boldsymbol{X}^\dagger \boldsymbol{Y}$ (Moore-Penrose pseudoinverse) has test error:

$$R(\widehat{\beta}) = \underbrace{\sigma^2 \cdot \operatorname{tr}(\boldsymbol{\Sigma}(\boldsymbol{X}\boldsymbol{X}^\top)^{-2}\boldsymbol{X}\boldsymbol{\Sigma}\boldsymbol{X}^\top/n)}_{\text{variance}} + \underbrace{\left\|(\boldsymbol{I} - \boldsymbol{X}^\dagger\boldsymbol{X})\beta^*\right\|_{\boldsymbol{\Sigma}}^2}_{\text{bias}}.$$

When $p/n \to \gamma > 1$, as $\gamma \to \infty$ (i.e., $p \gg n$), the variance term $\to 0$ and bias $\to 0$ under appropriate conditions on the spectrum of $\boldsymbol{\Sigma}$.

Intuition. In the overparameterized regime $p > n$, there are infinitely many interpolating solutions ($\boldsymbol{X}\beta = \boldsymbol{Y}$ exactly). Gradient descent selects the *minimum-norm* solution $\widehat{\beta} = \boldsymbol{X}^\dagger \boldsymbol{Y}$. When p is much larger than n, this solution distributes the signal very smoothly across many directions, avoiding any single-direction memorization of noise. The effective

dimension of the solution decreases, reducing variance — the opposite of what classical theory predicts.

Definition: Effective Rank (Bartlett et al., 2020)

Let $\lambda_1 \geq \lambda_2 \geq \cdots$ be the eigenvalues of $\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{x}\boldsymbol{x}^\top]$. The **effective rank** beyond the top k components is:

$$r_k(\boldsymbol{\Sigma}) = \frac{\sum_{i>k} \lambda_i}{\lambda_{k+1}}.$$

Benign overfitting requires r_k to be large: the tail of the spectrum is nearly isotropic (many eigenvalues of comparable size), so no single direction dominates. The minimum-norm interpolant spreads noise uniformly across the many small-eigenvalue directions, preventing any one direction from blowing up.

Implicit regularization. Gradient descent on the squared loss (without explicit regularization) converges to the minimum ℓ_2-norm interpolant — this is a form of *implicit* regularization. The noise is spread across *all* p parameters rather than concentrated in a few, keeping the variance per direction small when $p \gg n$.

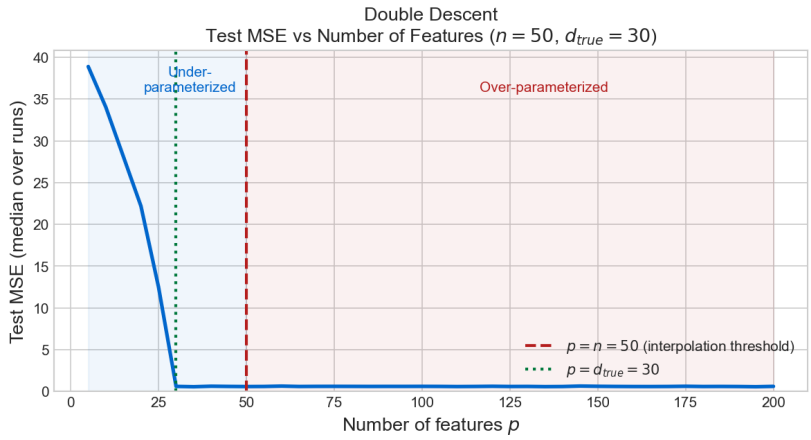


Figure 3.3: Double descent curve: test MSE vs. model dimension p for linear regression with $n = 50$ samples. The peak at $p = n$ is the interpolation threshold. For $p \gg n$, the minimum-norm interpolator achieves near-zero bias and declining variance.

Lecture 4

Neural Networks and Approximation Theory

Historical Note: From McCulloch-Pitts to Deep Learning

The artificial neuron was introduced by McCulloch and Pitts (1943) as a mathematical abstraction of biological neurons. Hebb's rule (1949) provided a local learning mechanism. Rosenblatt's perceptron (1958) added the first convergence theorem. The crucial backpropagation algorithm was described in Werbos's PhD thesis (1974) and popularized by Rumelhart, Hinton and Williams (1986). After a long winter following the XOR limitation (Minsky & Papert, 1969), deep learning exploded in 2012 when Krizhevsky, Sutskever and Hinton won ImageNet by a large margin using a deep convolutional network trained on GPUs. The 2024 Nobel Prize in Physics was awarded to Hopfield and Hinton for their foundational contributions to artificial neural networks.

4.1. The Perceptron

Definition: McCulloch-Pitts Neuron and Rosenblatt Perceptron

A single neuron computes $\hat{y} = \sigma(\boldsymbol{w}^\top \boldsymbol{x} + b)$ where $\boldsymbol{w} \in \mathbb{R}^p$ are weights, $b \in \mathbb{R}$ is a bias, and σ is a nonlinearity. The perceptron uses $\sigma = \text{sign}$:

$$\hat{y}_i = \text{sign}(\boldsymbol{w}^\top \boldsymbol{x}_i + b) \in \{-1, +1\}.$$

The **perceptron learning rule**: for each misclassified example (x_i, y_i):

$$\boldsymbol{w} \leftarrow \boldsymbol{w} + \eta y_i \boldsymbol{x}_i, \qquad b \leftarrow b + \eta y_i.$$

Perceptron Convergence Theorem (Rosenblatt, 1962; Block, 1962)

If there exists $\boldsymbol{w}^*$ with $\|\boldsymbol{w}^*\| = 1$ such that $y_i \boldsymbol{w}^{*\top} \boldsymbol{x}_i \geq \gamma > 0$ for all i (linearly separable with margin γ), and $\|\boldsymbol{x}_i\| \leq R$, then the perceptron algorithm makes at most $T \leq (R/\gamma)^2$ mistakes.

Proof

Track two quantities: (a) $\boldsymbol{w}^{(T)\top}\boldsymbol{w}^* \geq T\gamma$ (grows linearly with T, since each update adds $y_i\boldsymbol{x}_i$ and $\boldsymbol{w}^{*\top}(y_i\boldsymbol{x}_i) \geq \gamma$), and (b) $\left\|\boldsymbol{w}^{(T)}\right\|^2 \leq TR^2$ (grows at most linearly, since each update changes $\|\boldsymbol{w}\|^2$ by $\|y_i\boldsymbol{x}_i\|^2 \leq R^2$ when misclassified — here $y_i\boldsymbol{w}^\top\boldsymbol{x}_i \leq 0$ so the cross-term is non-positive). By Cauchy-Schwarz:

$$T\gamma \leq \boldsymbol{w}^{(T)\top}\boldsymbol{w}^* \leq \left\|\boldsymbol{w}^{(T)}\right\| \|\boldsymbol{w}^*\| \leq \sqrt{T}R.$$

Squaring: $T^2\gamma^2 \leq TR^2$, so $T \leq (R/\gamma)^2$. □

The XOR problem. XOR is not linearly separable: no hyperplane correctly classifies $(0,0) \to 0$, $(0,1) \to 1$, $(1,0) \to 1$, $(1,1) \to 0$. This fundamental limitation of single-layer perceptrons (Minsky & Papert, 1969) halted neural network research for over a decade, until multi-layer networks were developed.

4.2. Multi-Layer Perceptrons

Definition: Feedforward Neural Network (Multi-Layer Perceptron)

A depth-L network with input dimension $n_0 = p$ and layer widths $n_1, \ldots, n_L$:

$$\begin{aligned} \boldsymbol{z}^{(l)} &= \boldsymbol{W}^{(l)}\boldsymbol{a}^{(l-1)} + \boldsymbol{b}^{(l)}, \quad \boldsymbol{a}^{(l)} = \sigma(\boldsymbol{z}^{(l)}), \quad l = 1, \ldots, L-1, \\ \hat{y} &= \boldsymbol{W}^{(L)}\boldsymbol{a}^{(L-1)} + b^{(L)}, \end{aligned}$$

with $\boldsymbol{a}^{(0)} = \boldsymbol{x}$. The activation function σ is applied elementwise. The total parameter count is $|\theta| = \sum_{l=1}^{L} n_l(n_{l-1}+1)$.

Why depth? Without nonlinear activations: $\boldsymbol{W}^{(L)} \cdots \boldsymbol{W}^{(1)}\boldsymbol{x} = \widetilde{\boldsymbol{W}}\boldsymbol{x}$ — any L-layer linear network collapses to a single linear transformation. Nonlinearity is necessary for depth to add expressive power. Montufar et al. (2014) showed that a depth-L ReLU network with n units per layer creates up to $\Omega\big((n/p)^{p(L-1)}n^p\big)$ linear regions, growing exponentially in depth.

4.2.1. Activation Functions

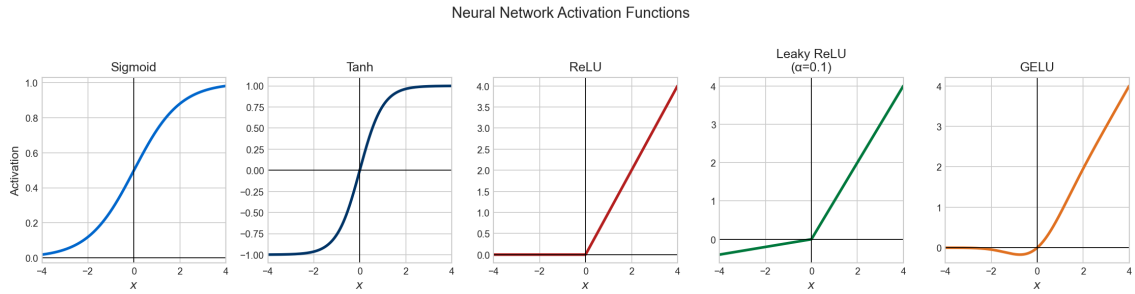


Figure 4.1: Five common activation functions. ReLU is the workhorse of modern deep learning. GELU (used in transformers) smoothly interpolates between a linear function and ReLU.

- **Sigmoid** $\sigma(z) = 1/(1+e^{-z})$: historical, used in output layers for binary classification. Saturates on both sides, causing vanishing gradients in deep networks.
- **Tanh** $\sigma(z) = (e^z - e^{-z})/(e^z + e^{-z})$: zero-centered sigmoid. Still saturates.
- **ReLU** $\sigma(z) = \max(0, z)$: piecewise linear, no saturation for $z > 0$. The de facto standard since 2012. Sparse activations (many zeros) have a regularization effect.
- **Dying ReLU problem**: neurons receiving mostly negative pre-activations have gradient zero and stop learning. Fix: Leaky ReLU $\max(\alpha z, z)$ with small $\alpha > 0$.
- **GELU** $\sigma(z) = z\Phi(z)$: smooth approximation to ReLU, used in BERT, GPT. No abrupt transition at zero.

4.3. Backpropagation

4.3.1. The Optimization Problem

Minimizing $\mathcal{L}(\theta) = \frac{1}{n}\sum_i \ell(y_i, \hat{y}_i(\theta))$ requires computing $\nabla_\theta \mathcal{L}$.

Why not numerical gradients? For P parameters, numerical differentiation requires $P+1$ forward passes (one per parameter plus one baseline). A network with 10^8 parameters (moderate by modern standards) would need 10^8 forward passes per gradient step — completely infeasible.

Why not direct chain rule expansion? The raw chain rule expands $\partial\mathcal{L}/\partial W^{(l)}$ as a product of Jacobians all the way from output to layer l. Each layer's computation is repeated fresh, leading to $O(L^2)$ redundant operations.

Backpropagation solves both problems: it computes all $|\theta|$ partial derivatives in exactly *one forward pass* (to cache intermediate values) plus *one backward pass* (to propagate error signals), for a total cost of $O(|\theta|)$.

4.3.2. The Forward Pass and Error Signals

Define the **error signal** (delta) at layer l:

$$\boldsymbol{\delta}^{(l)} \equiv \frac{\partial \mathcal{L}}{\partial \boldsymbol{z}^{(l)}} \in \mathbb{R}^{n_l}.$$

Once $\boldsymbol{\delta}^{(l)}$ is known, the weight gradient is:

$$\nabla_{\boldsymbol{W}^{(l)}} \mathcal{L} = \boldsymbol{\delta}^{(l)} (\boldsymbol{a}^{(l-1)})^\top.$$

4.3.3. The Backpropagation Recursion

Backpropagation (Werbos 1974; Rumelhart, Hinton, Williams 1986)

Initialize with the output layer: $\boldsymbol{\delta}^{(L)} = \nabla_{\hat{y}} \ell(y, \hat{y})$. For $l = L-1, L-2, \ldots, 1$:

$$\boldsymbol{\delta}^{(l)} = (\boldsymbol{W}^{(l+1)})^\top \boldsymbol{\delta}^{(l+1)} \odot \sigma'(\boldsymbol{z}^{(l)}),$$

where $\odot$ is elementwise multiplication. Weight and bias gradients:

$$\nabla_{\boldsymbol{W}^{(l)}} \mathcal{L} = \boldsymbol{\delta}^{(l)} (\boldsymbol{a}^{(l-1)})^\top, \qquad \nabla_{\boldsymbol{b}^{(l)}} \mathcal{L} = \boldsymbol{\delta}^{(l)}.$$

Derivation of recursion

By the chain rule, $\boldsymbol{\delta}^{(l)} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{z}^{(l)}}$. Since $\boldsymbol{z}^{(l+1)} = \boldsymbol{W}^{(l+1)}\boldsymbol{a}^{(l)} + \boldsymbol{b}^{(l+1)} = \boldsymbol{W}^{(l+1)}\sigma(\boldsymbol{z}^{(l)}) + \boldsymbol{b}^{(l+1)}$:

$$\frac{\partial \mathcal{L}}{\partial z_j^{(l)}} = \sum_k \frac{\partial \mathcal{L}}{\partial z_k^{(l+1)}} \frac{\partial z_k^{(l+1)}}{\partial z_j^{(l)}} = \sum_k \delta_k^{(l+1)} W_{kj}^{(l+1)} \sigma'(z_j^{(l)}).$$

In matrix form: $\boldsymbol{\delta}^{(l)} = (\boldsymbol{W}^{(l+1)})^\top \boldsymbol{\delta}^{(l+1)} \odot \sigma'(\boldsymbol{z}^{(l)})$. □

Key efficiency. The factor $(\boldsymbol{W}^{(l+1)})^\top \boldsymbol{\delta}^{(l+1)}$ is computed once and reused for all downstream computations. This is the dynamic programming insight: cache and reuse, rather than recompute.

4.3.4. Vanishing and Exploding Gradients

For sigmoid activations, $\sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq 1/4$. After L layers:

$$\left\|\boldsymbol{\delta}^{(1)}\right\| \leq \left\|\boldsymbol{W}^{(2)}\right\|^\top \cdot \frac{1}{4} \cdot \left\|\boldsymbol{W}^{(3)}\right\|^\top \cdot \frac{1}{4} \cdots \left\|\boldsymbol{\delta}^{(L)}\right\| \lesssim \left(\frac{\|W\|}{4}\right)^{L-1} \left\|\boldsymbol{\delta}^{(L)}\right\|.$$

If $\|W\| < 4$, gradients shrink exponentially — *vanishing gradients*. If $\|W\| > 4$, gradients grow exponentially — *exploding gradients*. Both problems plagued deep networks with sigmoid activations.

Solutions developed over time:

- ReLU: gradient is 1 for positive pre-activations, avoiding shrinkage in forward-active neurons.
- Careful initialization (Glorot & Bengio 2010, He et al. 2015): scale initial weights so that signal and gradient have approximately unit variance at each layer.
- Batch normalization (Ioffe & Szegedy 2015): normalize activations within each batch, stabilizing training.
- Residual connections / skip connections (He et al. 2016): $\boldsymbol{a}^{(l)} = \sigma(\boldsymbol{z}^{(l)}) + \boldsymbol{a}^{(l-2)}$ creates a gradient highway bypassing nonlinearities.

4.4. Approximation Theory

4.4.1. Universal Approximation

Universal Approximation Theorem (Cybenko 1989; Hornik, Stinchcombe & White 1989)

Let $\sigma : \mathbb{R} \to \mathbb{R}$ be continuous, bounded, and non-constant. For any $f \in C([0,1]^d)$ and $\varepsilon > 0$, there exist $N \in \mathbb{N}$, $\boldsymbol{w}_j \in \mathbb{R}^d$, $b_j, c_j \in \mathbb{R}$ such that:

$$\sup_{\boldsymbol{x} \in [0,1]^d} \left| f(\boldsymbol{x}) - \sum_{j=1}^N c_j \sigma(\boldsymbol{w}_j^\top \boldsymbol{x} + b_j) \right| < \varepsilon.$$

Constructive proof sketch for $d = 1$ (Cybenko, 1989)

We build an approximating network in four steps:

Step 1. Step function: $\sigma(w(x-c))$ with $c = -b/w$. As $w \to \infty$, $\sigma(w(x-c))$ becomes the indicator $\mathbb{1}[x \geq c]$, jumping from 0 to 1 at c.

Step 2. Bump indicator: $B(x) = \sigma(w(x-c_1)) - \sigma(w(x-c_2))$ on interval $[c_1, c_2]$. For $w \to \infty$, $B(x) \approx \mathbb{1}[x \in [c_1, c_2]]$ — a "bump" function equal to 1 on the interval.

Step 3. Piecewise constant: partition $[0,1]$ into intervals $[c_k, c_{k+1}]$ and approximate f by $f(x) \approx \sum_k f(x_k) B_k(x)$ — a piecewise constant function matching f at sample points.

Step 4. Single-layer network: $g(x) = \sum_k f(x_k)[\sigma(w(x-c_k)) - \sigma(w(x-c_{k+1}))]$. This is a linear combination of $2K$ sigmoid units and approximates f uniformly as $K \to \infty$ and $w \to \infty$.

□

Non-constructive proof — Hahn-Banach approach

Suppose no such network exists. Then the set $S = \{g = \sum_j c_j \sigma(\boldsymbol{w}_j^\top \boldsymbol{x} + b_j)\}$ is not dense in $C([0,1]^d)$ in the sup norm. By the Hahn-Banach theorem, there exists a nonzero continuous linear functional μ (a signed Borel measure) on $C([0,1]^d)$ such that $\int \sigma(\boldsymbol{w}^\top \boldsymbol{x} + b)\, d\mu(\boldsymbol{x}) = 0$ for all $\boldsymbol{w}, b$.
Define $F(\boldsymbol{w}) = \int e^{i\boldsymbol{w}^\top \boldsymbol{x}}\, d\mu(\boldsymbol{x})$ (Fourier transform of μ). The key is to show that $F \equiv 0$, which forces $\mu = 0$ by uniqueness of Fourier transform, a contradiction.
For any $\boldsymbol{w}_0, b_0$: the function $h_{\boldsymbol{w}_0, b_0}(t) = \int \sigma(t\boldsymbol{w}_0^\top \boldsymbol{x} + b_0)\, d\mu(\boldsymbol{x}) = 0$ for all t. Since σ is bounded and non-constant, it is not a polynomial, so its Fourier transform is not identically zero. The fact that $h \equiv 0$ for all $(\boldsymbol{w}_0, b_0)$ then implies $F \equiv 0$ on $\mathbb{R}^d$, contradiction. □

What UAT Does Not Say

- It is an *existence* theorem: the required N may be exponential in d.
- It gives no algorithm for finding the weights.
- It does not say that gradient descent will find the approximating network.
- It does not imply generalization — a network can perfectly approximate f^* on $[0,1]^d$ but overfit on finite samples.

4.4.2. Barron's Theorem: Quantitative, Dimension-Free Approximation

UAT is qualitative. Barron (1993) gave the first quantitative, dimension-free approximation rate.

Definition: Barron's Function Class

The **spectral norm** of $f : \mathbb{R}^d \to \mathbb{R}$ is:

$$C_f = \int_{\mathbb{R}^d} \|\boldsymbol{\omega}\|\, |\hat{f}(\boldsymbol{\omega})|\, d\boldsymbol{\omega},$$

where $\hat{f}(\boldsymbol{\omega}) = \int_{\mathbb{R}^d} e^{i\boldsymbol{\omega}^\top \boldsymbol{x}} f(\boldsymbol{x})\, d\boldsymbol{x}$ is the Fourier transform. Barron's class $\mathcal{B}(C) = \{f : C_f \leq C\}$ consists of functions with bounded spectral norm.

Barron's Theorem (1993)

For any $f \in \mathcal{B}(C)$, any distribution μ on a bounded domain, and any $N \geq 1$:

$$\inf_{f_N \in \mathcal{F}_N} \int (f(\boldsymbol{x}) - f_N(\boldsymbol{x}))^2 \, d\mu(\boldsymbol{x}) \leq \frac{C^2}{N},$$

where $\mathcal{F}_N$ is the class of single-hidden-layer networks with N sigmoid units.

Proof by probabilistic method

Write $f(\boldsymbol{x}) = \int \hat{f}(\boldsymbol{\omega}) e^{i\boldsymbol{\omega}^\top \boldsymbol{x}} \, d\boldsymbol{\omega}$. By the inverse Fourier representation (for the real part):

$$f(\boldsymbol{x}) = \int |\hat{f}(\boldsymbol{\omega})| \cdot \frac{\hat{f}(\boldsymbol{\omega})}{|\hat{f}(\boldsymbol{\omega})|} e^{i\boldsymbol{\omega}^\top \boldsymbol{x}} \, d\boldsymbol{\omega}.$$

This is an expectation: $f(\boldsymbol{x}) = C_f \mathbb{E}_{\boldsymbol{\omega} \sim \pi}[g_{\boldsymbol{\omega}}(\boldsymbol{x})]$, where π has density $|\hat{f}(\boldsymbol{\omega})|/C_f$ and $g_{\boldsymbol{\omega}}$ is a bounded single-neuron function. By the law of large numbers, the empirical average over N i.i.d. draws $\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_N \sim \pi$:

$$f_N(\boldsymbol{x}) = C_f \cdot \frac{1}{N} \sum_{j=1}^{N} g_{\boldsymbol{\omega}_j}(\boldsymbol{x})$$

has $\int (f - f_N)^2 \, d\mu \leq C_f^2/N$ in expectation (variance of empirical mean over N samples). Hence a good N-neuron network exists. □

Why is this remarkable?

- The rate $O(1/N)$ is *dimension-free*: it does not depend on d.
- Classical polynomial approximation in $\mathbb{R}^d$ with N terms achieves only $O(N^{-2k/d})$ for k-smooth functions, which deteriorates catastrophically in high dimensions (curse of dimensionality).
- Barron's class includes functions relevant to finance (option pricing formulae), physics, and signal processing, explaining why neural networks excel in these domains.

Example: Finance: Neural Networks and the Curse of Dimensionality

Pricing a European option on a basket of $d = 100$ underlyings requires evaluating a 100-dimensional integral. Monte Carlo methods converge at $O(1/\sqrt{N})$ regardless of d. A polynomial approximation of the payoff function in 100 dimensions requires millions of terms to achieve modest accuracy. Barron's theorem implies a neural network with $N = 1000$ units can approximate the payoff function with error $\leq C^2/1000$ — again independent of $d = 100$, provided the payoff lies in Barron's class (which exponential payoffs do).

4.5. The Lazy / NTK Regime and Benign Overfitting in Neural Networks

4.5.1. The Linear / Lazy Regime

In the massively overparameterized regime ($|\theta| \gg n$), neural networks can be analyzed as linear models. Consider the first-order Taylor expansion of the network around its random initialization θ_0:

$$f(\boldsymbol{x};\theta) \approx f(\boldsymbol{x};\theta_0) + \nabla_\theta f(\boldsymbol{x};\theta_0)^\top(\theta - \theta_0).$$

If the weights barely move from initialization ($\|\theta - \theta_0\| \ll 1$), this approximation is accurate globally. The network is effectively linear in $\theta - \theta_0$ with feature map $\boldsymbol{\phi}(\boldsymbol{x}) = \nabla_\theta f(\boldsymbol{x};\theta_0)$.

Definition: Neural Tangent Kernel (Jacot, Gabriel & Hongler, 2018)

The **Neural Tangent Kernel** (NTK) is:

$$\Theta(\boldsymbol{x},\boldsymbol{x}') = \nabla_\theta f(\boldsymbol{x};\theta_0)^\top \nabla_\theta f(\boldsymbol{x}';\theta_0).$$

In the infinite-width limit, gradient descent on the squared loss converges at rate $e^{-\lambda_{\min}(\Theta)t}$ and the trained network's predictions are:

$$\hat{\boldsymbol{Y}} = \boldsymbol{K}(\boldsymbol{K} + \lambda \boldsymbol{I})^{-1}\boldsymbol{Y},$$

where $\boldsymbol{K}_{ij} = \Theta(\boldsymbol{x}_i, \boldsymbol{x}_j)$. This is *exactly* kernel regression with the NTK.

Implicit regularization. In the NTK regime, gradient descent with sufficiently small learning rate converges to the minimum ℓ_2-norm interpolating solution in the feature space. This is the minimum-norm interpolant, which tends to smooth predictions between data points — providing implicit regularization without any explicit regularization term.

4.5.2. Stochastic Gradient Descent

In practice, networks are trained with stochastic gradient descent (SGD) or Adam:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\mathcal{B}}(\theta),$$

where $\mathcal{B}$ is a mini-batch of size $B \ll n$. SGD provides three benefits over full-batch gradient descent:

1. **Speed**: one epoch requires n/B gradient steps at cost $O(B|\theta|)$ each, vs. one gradient step at cost $O(n|\theta|)$.
2. **Implicit regularization**: the gradient noise helps escape sharp minima and find flatter, more generalizable solutions (Hochreiter & Schmidhuber, 1997; Keskar et al., 2017).
3. **Nonconvex navigation**: gradient noise can help escape saddle points, which proliferate in the loss landscapes of deep networks.

Example: Application: Image Recognition (MNIST and beyond)

The MNIST handwritten digit dataset (60,000 training images, 28×28 pixels, 10 classes) was a standard benchmark for neural networks throughout the 1990s-2000s. A 3-layer ReLU network achieves $> 99\%$ test accuracy. The 2012 ImageNet competition ($1.2M$ images, 1000 classes) was won by AlexNet (8 layers, 60M parameters) with 15.3% top-5 error — more than 10 percentage points better than the previous best. This marked the beginning of the deep learning era. Modern networks (ResNet-50,

ViT) achieve $< 3\%$ error on ImageNet.

Lecture 5

Variational Inference, VAEs, and Langevin Dynamics

Historical Note: The Origins of Variational Inference

The idea of approximating intractable distributions by optimization has roots in statistical physics (mean-field theory for the Ising model, Peierls 1936). The machine learning formalization came from Jordan, Ghahramani, Jaakkola and Saul (1999), who introduced variational inference as a scalable alternative to MCMC. Blei, Kucukelbir and McAuliffe (2017) provided the modern algorithmic treatment with automatic differentiation. The variational autoencoder (Kingma & Welling, 2013; Rezende, Mohamed & Wierstra, 2014) brought generative modeling with latent variables to the deep learning era, winning the 2023 ACM AAAI Allen Newell Award.

5.1. From Prediction to Generative Modeling

So far we have learned functions $f : \mathcal{X} \to \mathcal{Y}$ — point predictors. Generative probabilistic modeling has a different goal: learn the full joint distribution $p(x)$ or $p(x, z)$.

Why generative models?

1. **Density estimation**: understand the data distribution itself (anomaly detection, compression).
2. **Generation**: sample new $x \sim p(x)$ (image synthesis, drug discovery).
3. **Representation learning**: encode x into a low-dimensional latent z capturing semantically meaningful variation.
4. **Causal reasoning**: $p(\mathrm{do}(z = z_0))$ requires a model of the joint (x, z).

Latent variable models. Posit hidden (latent) variables $z \in \mathbb{R}^k$ governing the observed data x:

$$z \sim p(z), \qquad x \mid z \sim p_\theta(x \mid z).$$

The marginal likelihood is:

$$p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, dz.$$

For flexible p_θ (e.g., the decoder is a neural network), this integral is intractable. Similarly, the posterior $p_\theta(z \mid x) = p_\theta(x \mid z)p(z)/p_\theta(x)$ is intractable because $p_\theta(x)$ is.

5.2. Information Theory: Entropy and KL Divergence

Definition: Shannon Entropy

The **entropy** of a distribution p over $\mathcal{X}$ is:

$$H(p) = -\mathbb{E}_p[\log p(X)] = -\sum_{x \in \mathcal{X}} p(x) \log p(x) \quad \text{(discrete)}.$$

For continuous distributions: $h(p) = -\int p(x) \log p(x)\, dx$ (differential entropy). Entropy measures *average uncertainty* or *average surprise* in p. For a fair coin, $H = \log 2$ bits; for a loaded coin with $p(H) = 0.99$, H is small.

Properties of Entropy

(i) $H(p) \geq 0$ for discrete distributions.
(ii) $H(p) \leq \log |\mathcal{X}|$, with equality iff p is uniform (maximum entropy principle).
(iii) For Gaussian $X \sim \mathcal{N}(\mu, \sigma^2)$: $h(X) = \frac{1}{2} \log(2\pi e \sigma^2)$.

Definition: Kullback–Leibler Divergence

The **KL divergence** from q to p is:

$$\mathrm{KL}(p \| q) = \mathbb{E}_p\left[\log \frac{p(X)}{q(X)}\right] = \int p(x) \log \frac{p(x)}{q(x)}\, dx.$$

It measures the information lost when using q to approximate p.

Gibbs' Inequality: Non-Negativity of KL

$\mathrm{KL}(p \| q) \geq 0$ for all distributions p, q, with equality iff $p = q$ almost everywhere.

Proof

By Jensen's inequality and the concavity of log:

$$-\mathrm{KL}(p \| q) = \mathbb{E}_p\left[\log \frac{q(X)}{p(X)}\right] \leq \log \mathbb{E}_p\left[\frac{q(X)}{p(X)}\right] = \log \int q(x)\, dx = \log 1 = 0.$$

Equality in Jensen holds iff $q(X)/p(X)$ is constant p-a.s., i.e., $q = cp$ for some constant c; normalization forces $c = 1$. □

5.2.1. KL is Not Symmetric: Forward vs. Reverse KL

In variational inference we must choose which direction of KL to minimize.

- **Forward KL**: $\min_q \mathrm{KL}(p \| q) = \mathbb{E}_p[\log p(X)/q(X)]$. Penalizes $q(x) \approx 0$ when $p(x) > 0$ heavily. Forces q to cover all modes of p (*mass-covering*, *mean-seeking*). Intractable

because it requires samples from p.

- **Reverse KL**: $\min_q \mathrm{KL}(q\|p) = \mathbb{E}_q[\log q(X)/p(X)]$. Penalizes $q(x) > 0$ when $p(x) \approx 0$ heavily. Forces q to concentrate on the high-probability regions of p (*mode-seeking*). **This is what VI minimizes** — tractable because samples from q are available.

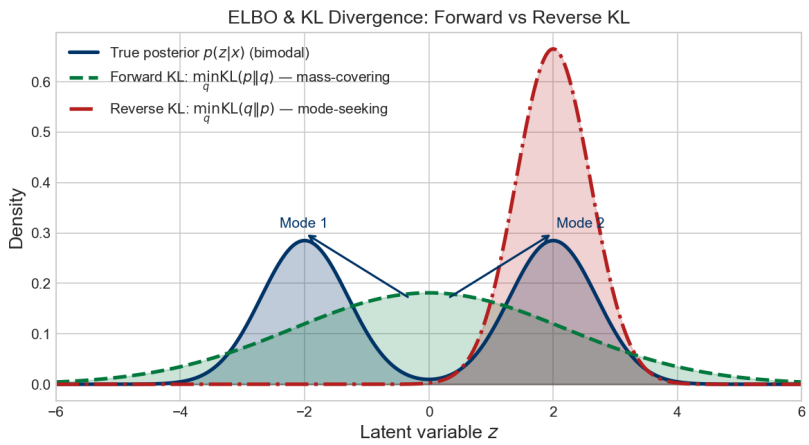


Figure 5.1: Forward vs. reverse KL on a bimodal target $p(z|x)$. Forward KL produces a broad, mass-covering approximation. Reverse KL produces a narrow, mode-seeking approximation (mode collapse). Both are unimodal Gaussians, so neither perfectly captures the bimodal structure.

Mode collapse. In variational inference, reverse KL is minimized, causing q to prefer one mode of p over another. This is the fundamental limitation of mean-field VI: it systematically underestimates uncertainty.

5.3. The Evidence Lower Bound (ELBO)

ELBO Decomposition (Jordan et al., 1999)

For any distribution $q(z)$:

$$\log p_\theta(x) = \underbrace{\mathbb{E}_{q(z)}\left[\log \frac{p_\theta(x,z)}{q(z)}\right]}_{\mathcal{L}(\theta,q;\,x)} + \underbrace{\mathrm{KL}(q(z)\|p_\theta(z|x))}_{\geq 0}.$$

Hence $\mathcal{L}(\theta, q; x) \leq \log p_\theta(x)$, with equality iff $q(z) = p_\theta(z|x)$.

Proof

$$\begin{aligned}
\log p_\theta(x) &= \log \int p_\theta(x,z)\,dz = \log \int q(z)\frac{p_\theta(x,z)}{q(z)}\,dz \\
&\geq \int q(z) \log \frac{p_\theta(x,z)}{q(z)}\,dz \quad \text{(Jensen's inequality, log is concave)} \\
&= \mathcal{L}(\theta, q; x).
\end{aligned}$$

The gap is $\log p_\theta(x) - \mathcal{L} = \mathrm{KL}(q \| p_\theta(\cdot|x)) \geq 0$ by Gibbs' inequality. □

Alternative derivation via Bayes' theorem (from hand notes)

Start from $\log p_\theta(x) = \int q(z) \log p_\theta(x)\, dz$ (since $\log p_\theta(x)$ does not depend on z). Use Bayes: $p_\theta(x) = p_\theta(x,z)/p_\theta(z|x)$:

$$\begin{aligned}\log p_\theta(x) &= \int q(z) \log \frac{p_\theta(x,z)}{p_\theta(z|x)}\, dz = \int q(z) \log \frac{p_\theta(x,z)}{q(z)} \cdot \frac{q(z)}{p_\theta(z|x)}\, dz \\ &= \underbrace{\int q(z) \log \frac{p_\theta(x,z)}{q(z)}\, dz}_{\mathcal{L}(\theta,q;x)} + \underbrace{\int q(z) \log \frac{q(z)}{p_\theta(z|x)}\, dz}_{\mathrm{KL}(q\|p_\theta(\cdot|x))\geq 0}.\end{aligned}$$

This derivation is exact (no Jensen needed) and makes the gap $= \mathrm{KL}(q\|p_\theta(\cdot|x))$ completely transparent. The ELBO equals $\log p_\theta(x)$ precisely when $q = p_\theta(\cdot|x)$. □

Two equivalent forms of the ELBO:

$$\begin{aligned}\mathcal{L}(\theta, q; x) &= \mathbb{E}_{q(z)}[\log p_\theta(x,z)] - \mathbb{E}_{q(z)}[\log q(z)] \\ &= \mathbb{E}_{q(z)}[\log p_\theta(x|z)] + \mathbb{E}_{q(z)}[\log p(z)] - \mathbb{E}_{q(z)}[\log q(z)] \\ &= \underbrace{\mathbb{E}_{q(z)}[\log p_\theta(x|z)]}_{\text{reconstruction}} - \underbrace{\mathrm{KL}(q(z)\|p(z))}_{\text{regularization}}.\end{aligned} \tag{5.1}$$

The reconstruction term rewards q for placing mass on z values that explain x well. The KL term penalizes q for deviating from the prior — a regularization toward the prior belief.

Interpretation from physics. In statistical physics, $p(x,z) \propto e^{-E(x,z)}$ where E is the energy. The ELBO equals $H(q) + \mathbb{E}_q[-E(x,z)]$: entropy of q plus expected negative energy. Maximizing ELBO is equivalent to finding q that minimizes the *variational free energy* — the physics analog of finding thermal equilibrium.

5.4. Coordinate Ascent Variational Inference (CAVI)

Definition: Mean-Field Approximation

The **mean-field** family factorizes q fully:

$$q(z_1, \ldots, z_k) = \prod_{j=1}^{k} q_j(z_j).$$

All latent dimensions are assumed independent under the approximation.

Under mean-field, the optimal factor $q_j^*(z_j)$ that maximizes ELBO while holding all other $q_{j'}$ fixed is:

Optimal Mean-Field Factor

$$q_j^*(z_j) \propto \exp\Big(\mathbb{E}_{q_{-j}}[\log p(x,z)]\Big),$$

where $\mathbb{E}_{q_{-j}}$ denotes expectation over all variables except z_j.

Proof

Write $\mathcal{L} = \mathbb{E}_q[\log p(x,z)] - \mathbb{E}_q[\log q(z)] = \mathbb{E}_{q_j}\mathbb{E}_{q_{-j}}[\log p(x,z)] - \sum_{j'} \mathbb{E}_{q_{j'}}[\log q_{j'}(z_{j'})]$. For a fixed j, only the terms involving q_j matter:

$$\mathcal{L}\big|_{q_j} = \mathbb{E}_{q_j}[\underbrace{\mathbb{E}_{q_{-j}}[\log p(x,z)]}_{=:\log \tilde{p}_j(z_j)}] - \mathbb{E}_{q_j}[\log q_j(z_j)] + \text{const} = -\operatorname{KL}(q_j \| \tilde{p}_j) + \text{const}.$$

This is maximized when $q_j = \tilde{p}_j$, i.e., $q_j^*(z_j) \propto \exp(\mathbb{E}_{q_{-j}}[\log p(x,z)])$. □

CAVI algorithm: initialize all q_j arbitrarily (e.g., random distributions in the family), then cycle through $j = 1, \ldots, k$, updating each q_j to its optimum while holding others fixed. Repeat until the ELBO converges. This is guaranteed to converge to a local maximum (the ELBO is a non-concave function of all q_j jointly, but each coordinate problem is concave).

Example: CAVI for a Gaussian Model

Consider $z \sim \mathcal{N}(0,1)$, $x \mid z \sim \mathcal{N}(z, \sigma^2)$. Mean-field uses $q(z) = \mathcal{N}(\mu, v)$. The ELBO is:

$$\mathcal{L}(\mu, v) = -\frac{1}{2v} - \frac{(\mu - x)^2 + v}{2\sigma^2} - \frac{1}{2}\log v + \text{const}.$$

Setting derivatives to zero: $\hat{\mu} = x/(1+\sigma^2)$, $\hat{v} = \sigma^2/(1+\sigma^2)$. This matches the exact posterior $p(z|x) = \mathcal{N}(x/(1+\sigma^2), \sigma^2/(1+\sigma^2))$ — mean-field is exact here because the true posterior is already Gaussian and thus factorizes trivially. For multimodal or strongly coupled posteriors, mean-field gives only an approximation.

5.5. Variational Autoencoders

Definition: VAE Generative Model (Kingma & Welling, 2013)

$$\begin{aligned} p(z) &= \mathcal{N}(\mathbf{0}, \boldsymbol{I}), && \text{prior}, \\ p_\theta(x \mid z) &= \mathcal{N}(\mu_\theta(z), \sigma^2 \boldsymbol{I}), && \text{decoder (neural network } \mu_\theta), \\ q_\phi(z \mid x) &= \mathcal{N}(\mu_\phi(x), \operatorname{diag}(\sigma_\phi^2(x))), && \text{encoder (neural network } \mu_\phi, \sigma_\phi). \end{aligned}$$

The ELBO for the VAE decomposes as:

$$\mathcal{L}(\theta, \phi; x) = \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]}_{\text{reconstruction loss}} - \underbrace{\operatorname{KL}(q_\phi(z|x) \| \mathcal{N}(\mathbf{0}, \boldsymbol{I}))}_{\text{regularization}}.$$

The KL term has a closed form for diagonal Gaussians:

Closed-Form KL for Diagonal Gaussians

With $q_\phi(z|x) = \mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2))$ and $p(z) = \mathcal{N}(\mathbf{0}, \boldsymbol{I})$:

$$\mathrm{KL}(q_\phi \| \mathcal{N}(\mathbf{0}, \boldsymbol{I})) = \frac{1}{2}\sum_{j=1}^{k}(\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2).$$

Proof

For univariate Gaussians $q = \mathcal{N}(\mu, \sigma^2)$, $p = \mathcal{N}(0, 1)$:

$$\begin{aligned}\mathrm{KL}(q\|p) &= \int q(z) \log \frac{q(z)}{p(z)}\, dz = \mathbb{E}_q\left[-\frac{(z-\mu)^2}{2\sigma^2} + \frac{z^2}{2} + \log\frac{1}{\sigma}\right] \\ &= -\frac{1}{2} + \frac{\mu^2 + \sigma^2}{2} - \log\sigma = \frac{1}{2}(\mu^2 + \sigma^2 - 1 - \log\sigma^2).\end{aligned}$$

Sum over k independent dimensions. □

5.5.1. The Reparameterization Trick

To train by gradient descent, we need $\nabla_\phi \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$. The expectation is over q_ϕ, which depends on ϕ — differentiating through the sampling operation is not directly possible.

Reparameterization Trick (Kingma & Welling, 2013)

Write z as a deterministic transformation of ϕ and independent noise $\xi \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$:

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \xi.$$

Then:

$$\nabla_\phi \mathbb{E}_{q_\phi}[f(z)] = \nabla_\phi \mathbb{E}_{\xi\sim\mathcal{N}(\mathbf{0},\boldsymbol{I})}[f(\mu_\phi(x) + \sigma_\phi(x) \odot \xi)] = \mathbb{E}_\xi[\nabla_\phi f(\mu_\phi + \sigma_\phi \odot \xi)].$$

The gradient passes through the deterministic function of ϕ, making backpropagation applicable.

Proof

Since ξ is independent of ϕ, the exchange of gradient and expectation is valid by dominated convergence (assuming the gradient and expectation are bounded). The gradient with respect to ϕ acts only on the deterministic function $z(\phi) = \mu_\phi + \sigma_\phi \odot \xi$. □

5.5.2. Geometry of the VAE Latent Space

A standard autoencoder (AE) maps each x to a point $z = e_\phi(x)$ in the latent space. The latent space may have "holes" — regions z that decode to nonsense because no training data was encoded there. Interpolating between two codes z_1 and z_2 passes through uninformative regions.

The VAE encodes to a *distribution* $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \sigma_\phi^2(x)\boldsymbol{I})$. The KL term $\mathrm{KL}(q_\phi\|\mathcal{N}(\mathbf{0}, \boldsymbol{I}))$

pushes all encodings toward the unit Gaussian, forcing the latent space to be *dense and continuous*. As a consequence, decoding any point in $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$ produces a realistic sample, and linear interpolation between two latent codes produces smooth transitions.

Example: Application: Drug Discovery with VAEs

Gómez-Bombarelli et al. (2018) applied a VAE to molecular graphs encoded as SMILES strings. The encoder maps molecules to a continuous latent space; the decoder generates new SMILES from latent codes. Bayesian optimization in the smooth latent space explores the chemical space efficiently, discovering molecules with desired properties (drug-likeness, binding affinity). This approach, published in *ACS Central Science*, is now a standard method in computational drug discovery.

5.6. Langevin Dynamics

Langevin dynamics provides an MCMC approach to sampling that uses gradient information — a bridge between optimization and sampling.

Definition: Langevin Dynamics

Given a target $p(x) \propto e^{-U(x)}$ (with energy function U), the overdamped Langevin SDE is:

$$dx_t = -\nabla U(x_t)\, dt + \sqrt{2}\, dW_t,$$

where W_t is a standard Wiener process. The stationary distribution is $p(x) \propto e^{-U(x)}$. The discretized (unadjusted) Langevin algorithm (ULA) is:

$$x_{t+1} = x_t + \eta\, \nabla_x \log p(x_t) + \sqrt{2\eta}\, \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}).$$

The update $\eta \nabla_x \log p(x_t)$ is a gradient step toward the high-density regions of p; the noise $\sqrt{2\eta}\varepsilon_t$ ensures exploration and prevents convergence to a mode. In the limit $\eta \to 0$, ULA converges to the continuous Langevin SDE whose stationary distribution is p.

ULA Convergence (Dalalyan, 2017)

Assume $p(x) \propto e^{-U(x)}$ with U L-smooth and m-strongly convex. Then ULA with step size $\eta \leq 2/(m + L)$ produces iterates satisfying:

$$W_2(p_{x_T}, p) \leq (1 - m\eta)^T W_2(p_{x_0}, p) + C\sqrt{d\eta},$$

where W_2 is the Wasserstein-2 distance and d is the dimension. To reach $W_2 \leq \varepsilon$: choose $\eta = O(\varepsilon^2/d)$ and $T = O((L/m)\log(1/\varepsilon))$.

Connection to score matching and generative models. If we can estimate $s(x) = \nabla_x \log p(x)$ (the *score function*) — without knowing $p(x)$ itself — we can run Langevin dynamics to sample from p. This is the key insight behind score-based generative models: train a neural network $s_\phi(x) \approx \nabla_x \log p_{\text{data}}(x)$, then run ULA with the learned score to generate new samples. The bridge from Langevin dynamics (Ch. 5) to DDPM and Flow Matching (Ch. 6) runs through this idea.

Lecture 6

Diffusion Models and Flow Matching

Historical Note: The Road to Modern Generative Models

Generative Adversarial Networks (Goodfellow et al., 2014) initiated modern deep generative modeling but suffered from training instability and mode collapse. Variational autoencoders (Kingma & Welling, 2013) were stable but produced blurry samples. Normalizing flows (Rezende & Mohamed, 2015; Dinh et al., 2017) were exact but limited in architecture. The breakthrough came with score matching (Hyvärinen, 2005) and Denoising Score Matching (Vincent, 2011), which Song and Ermon (2019) scaled to images via annealed Langevin dynamics. Ho et al. (2020) showed that the same framework, cast as a hierarchical VAE (DDPM), achieved unprecedented sample quality. Song et al. (2021) unified the framework via stochastic differential equations. Flow Matching (Lipman et al., 2022; Albergo & Vanden-Eijnden, 2022) then provided an ODE-based perspective that is simpler to train and achieves competitive or superior results.

6.1. Distances Between Distributions

Before studying generative models, we need ways to measure the discrepancy between the model distribution p_θ and the data distribution p_{data}.

- **KL divergence**: $\mathrm{KL}(p_{\text{data}} \| p_\theta)$. Used in maximum likelihood (MLE). Requires evaluating $p_\theta(x)$, which is intractable for many models.
- **Wasserstein-1 distance**: $W_1(p, q) = \sup_{\|f\|_L \leq 1} \mathbb{E}_p[f(X)] - \mathbb{E}_q[f(X)]$. Has geometric meaning (earth mover's distance). Smooth even when supports don't overlap. Used in Wasserstein GANs.
- **Maximum Mean Discrepancy (MMD)**: $\mathrm{MMD}^2(p, q) = \|\mathbb{E}_p[\phi(X)] - \mathbb{E}_q[\phi(X)]\|_{\mathcal{H}}^2$ in a reproducing kernel Hilbert space.
- **Score-based objective**: instead of measuring $\mathrm{KL}(p_{\text{data}} \| p_\theta)$, learn $\nabla_x \log p_{\text{data}}(x)$ and use it for sampling. Bypasses the intractability of $p_\theta(x)$.

6.2. Denoising Diffusion Probabilistic Models (DDPM)

6.2.1. From VAE to DDPM

DDPM (Ho et al., 2020) is a hierarchical VAE where the encoder is not learned but fixed as a Gaussian noising chain.

Standard VAE. Encoder $q_\phi(z|x)$ and decoder $p_\theta(x|z)$ are both learned jointly.

DDPM. The encoder is a fixed Markov chain that gradually destroys the data with Gaussian noise over T steps:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t;\ \sqrt{\alpha_t}\, x_{t-1},\ (1-\alpha_t)\boldsymbol{I}), \quad t = 1, \ldots, T,$$

with $\alpha_t = 1 - \beta_t$ and noise schedule $0 < \beta_1 < \cdots < \beta_T < 1$. Only the decoder (denoiser) is learned.

6.2.2. The Forward Process

Closed-Form Marginal of Forward Process

Define $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. Then:

$$q(x_t|x_0) = \mathcal{N}(x_t;\ \sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t)\boldsymbol{I}).$$

Equivalently: $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$, where $\varepsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$.

Proof

By induction. For $t = 1$: $q(x_1|x_0) = \mathcal{N}(\sqrt{\alpha_1}x_0, (1-\alpha_1)\boldsymbol{I})$, consistent with $\bar{\alpha}_1 = \alpha_1$. Inductive step: if $x_{t-1}|x_0 \sim \mathcal{N}(\sqrt{\bar{\alpha}_{t-1}}x_0, (1-\bar{\alpha}_{t-1})\boldsymbol{I})$ and $x_t|x_{t-1} \sim \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)\boldsymbol{I})$, then by the Gaussian sum formula:

$$x_t|x_0 \sim \mathcal{N}(\sqrt{\alpha_t}\sqrt{\bar{\alpha}_{t-1}}x_0,\ \alpha_t(1-\bar{\alpha}_{t-1})\boldsymbol{I} + (1-\alpha_t)\boldsymbol{I}) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\boldsymbol{I}).$$

□

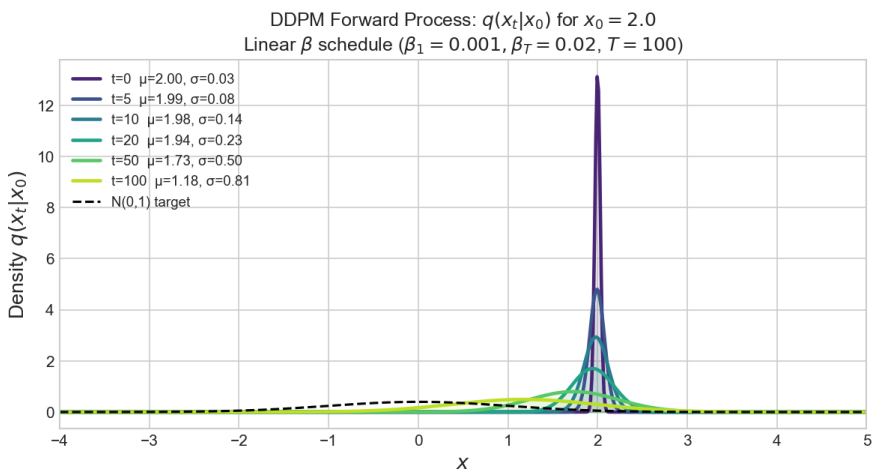


Figure 6.1: Forward diffusion process: the distribution $q(x_t|x_0)$ for $x_0 = 2$ becomes progressively broader and centered on zero. By $t = T$, the signal is destroyed and $q(x_T|x_0) \approx \mathcal{N}(0, 1)$.

As $T \to \infty$ with appropriate schedule, $\bar{\alpha}_T \to 0$ and $q(x_T|x_0) \approx \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ — the signal is completely destroyed. This provides the noise prior from which we start generation.

6.2.3. The Reverse Process

The reverse process is parameterized as:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1};\ \mu_\theta(x_t, t),\ \sigma_t^2 \boldsymbol{I}).$$

True Reverse Posterior is Tractable Conditioned on x_0

Conditioning on x_0 (available at training time), the posterior $q(x_{t-1}|x_t, x_0)$ is Gaussian:

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1};\ \tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t \boldsymbol{I}),$$

where:

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t, \qquad \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t.$$

Proof

Apply Bayes' theorem: $q(x_{t-1}|x_t, x_0) \propto q(x_t|x_{t-1}) q(x_{t-1}|x_0)$. Both are Gaussians, so their product (in the exponent) is a quadratic in x_{t-1}, giving a Gaussian. Completing the square yields $\tilde{\mu}_t$ and $\tilde{\beta}_t$. □

6.2.4. Training: The ELBO and Noise Prediction

The full DDPM ELBO decomposes into T KL terms:

$$-\log p_\theta(x_0) \le \underbrace{D_{\mathrm{KL}}(q(x_T|x_0)\|p(x_T))}_{L_T} + \sum_{t=2}^{T} \underbrace{D_{\mathrm{KL}}(q(x_{t-1}|x_t, x_0)\|p_\theta(x_{t-1}|x_t))}_{L_{t-1}} + \underbrace{-\log p_\theta(x_0|x_1)}_{L_0}.$$

$L_T \approx 0$ if T is large enough (fixed forward process). Each L_{t-1} compares two Gaussians, giving a closed-form MSE loss on the means.

Noise prediction parameterization. Express the mean via a noise-predicting network $\varepsilon_\theta(x_t, t)$:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(x_t, t)\right).$$

Substituting into the KL terms and simplifying (Ho et al. 2020 show the reweighting):

DDPM Simplified Training Loss

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}[1,T],\, x_0\sim p_{\text{data}},\, \varepsilon\sim\mathcal{N}(\mathbf{0},\boldsymbol{I})}\left[\left\|\varepsilon - \varepsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon,\ t)\right\|^2\right].$$

Intuition. Add noise ε to x_0 at level t, forming the noisy image x_t. Train ε_θ to predict the noise that was added. At inference, estimate the noise in x_t, use it to reconstruct a cleaner estimate, add a small amount of fresh noise, and repeat for $t = T, T-1, \ldots, 1$.

6.3. Score Matching

6.3.1. The Score Function

Definition: Score Function

The **score function** of $p(x)$ is:

$$s(x) = \nabla_x \log p(x).$$

It is a vector field on $\mathcal{X}$ pointing in the direction of steepest increase of the log-density.

For energy-based models $p_\phi(x) \propto e^{-E_\phi(x)}$, the intractable partition function cancels:

$$\nabla_x \log p_\phi(x) = -\nabla_x E_\phi(x).$$

We can estimate s without ever computing $p_\phi(x)$.

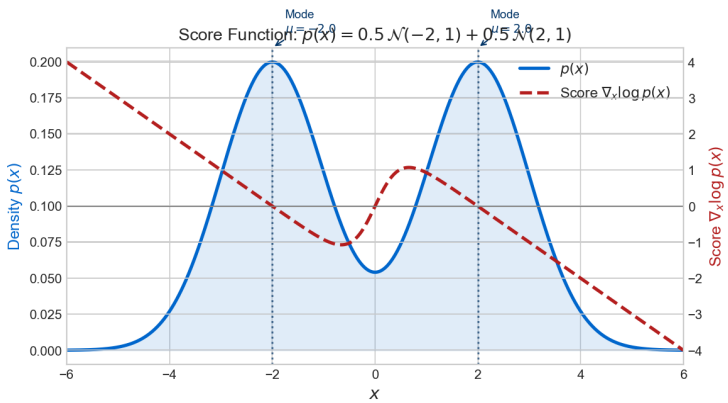


Figure 6.2: Score function of a bimodal distribution $p(x) = 0.5\mathcal{N}(-2, 1) + 0.5\mathcal{N}(2, 1)$. The score $\nabla_x \log p(x)$ (red) points toward the two modes, with a discontinuity at the saddle between modes. Langevin dynamics following this vector field converges to samples from p.

6.3.2. Score Matching (Hyvärinen, 2005)

Score Matching Objective

The **naive** score matching objective:

$$\mathcal{L}_{\mathrm{SM}}(\phi) = \frac{1}{2}\mathbb{E}_{p_{\mathrm{data}}}\left[\|s_\phi(x) - \nabla_x \log p_{\mathrm{data}}(x)\|^2\right]$$

is intractable (requires the true score). Hyvärinen (2005) showed it equals — up to a constant — the tractable:

$$\tilde{\mathcal{L}}_{\mathrm{SM}}(\phi) = \mathbb{E}_{p_{\mathrm{data}}}\left[\mathrm{tr}(\nabla_x s_\phi(x)) + \frac{1}{2}\|s_\phi(x)\|^2\right].$$

Both have the same minimizer $s^*_\phi = \nabla_x \log p_{\mathrm{data}}$.

Proof

Expand $\mathcal{L}_{\text{SM}}$ and integrate by parts:

$$\mathcal{L}_{\text{SM}} = \frac{1}{2}\mathbb{E}[\|s_\phi\|^2] + \frac{1}{2}\mathbb{E}[\|\nabla \log p\|^2] - \mathbb{E}[s_\phi^\top \nabla \log p].$$

The last term: $-\mathbb{E}[s_\phi^\top \nabla \log p] = -\int s_\phi(x)^\top \nabla p(x)\, dx$. By parts (assuming p decays at boundaries): $= \int \operatorname{tr}(\nabla_x s_\phi(x)) p(x)\, dx = \mathbb{E}[\operatorname{tr}(\nabla_x s_\phi(x))]$. So $\mathcal{L}_{\text{SM}} = \tilde{\mathcal{L}}_{\text{SM}} + \text{const}$. □

The Jacobian trace $\operatorname{tr}(\nabla_x s_\phi)$ costs $O(D^2)$ to compute via forward-mode AD — expensive in high dimensions.

6.3.3. Denoising Score Matching (Vincent, 2011)

DSM Avoids the Jacobian

Perturb x with Gaussian noise: $\tilde{x} = x + \sigma\varepsilon$, $\varepsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$. The conditional score has a closed form:

$$\nabla_{\tilde{x}} \log p_\sigma(\tilde{x}|x) = -\frac{\varepsilon}{\sigma} = \frac{x - \tilde{x}}{\sigma^2}.$$

The DSM loss is:

$$\mathcal{L}_{\text{DSM}}(\phi; \sigma) = \frac{1}{2}\mathbb{E}_{\tilde{x},\varepsilon}\left[\left\| s_\phi(\tilde{x}; \sigma) + \frac{\varepsilon}{\sigma}\right\|^2\right].$$

Theorem (Vincent, 2011): $\mathcal{L}_{\text{DSM}} = \mathcal{L}_{\text{SM}} + \text{const}$. Same minimizer $s_\phi^* = \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$.

Proof

$\mathcal{L}_{\text{DSM}} = \frac{1}{2}\mathbb{E}\,\|s_\phi(\tilde{x}) - \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}|\, x)\|^2$. Expand and integrate out x using $p_\sigma(\tilde{x}) = \int p_\sigma(\tilde{x}|x) p_{\text{data}}(x)\, dx$; the cross-term equals $\mathcal{L}_{\text{SM}}$. □

Connection to DDPM: the score and noise predictor are related by $s^*(x_t, t) = -\varepsilon^*(x_t, t)/\sqrt{1 - \bar{\alpha}_t}$ via Tweedie's formula. Score matching, DSM, and DDPM noise prediction are thus three equivalent parameterizations of the same underlying object.

6.4. Flow Matching

6.4.1. The ODE Perspective

Definition: Continuous Normalizing Flow / Flow Matching

Find a time-varying velocity field $v_\phi : \mathcal{X} \times [0, 1] \to \mathcal{X}$ such that integrating:

$$\frac{dx(t)}{dt} = v_\phi(x(t), t), \qquad t \in [0, 1],$$

transforms samples from $p_0 = \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ at $t = 0$ into samples from $p_1 = p_{\text{data}}$ at $t = 1$.

The induced density satisfies the **continuity equation**:

$$\frac{\partial p_t}{\partial t} + \nabla_x \cdot (p_t v_t) = 0.$$

Any v_t satisfying this equation correctly transports mass from p_0 to p_1.

Advantages of the ODE perspective over SDE (diffusion):

- Continuous, smooth trajectories (no stochastic noise at inference).
- Can be integrated with high-order ODE solvers in far fewer steps (10–20 vs. 1000 for DDPM).
- Exact likelihood computation via the instantaneous change-of-variables formula.
- Direct connection to optimal transport.

6.4.2. Conditional Flow Matching (CFM)

The marginal velocity $v_t(x) = \mathbb{E}[\dot{x}(t)|x(t) = x]$ is intractable because it requires knowing p_t everywhere. The **conditional flow matching** trick avoids this.

CFM Theorem (Lipman et al., 2022; Albergo & Vanden-Eijnden, 2022)

Let $z = x_1 \sim p_{\text{data}}$ and define conditional paths $p_t(x|z)$ with closed-form conditional velocity $v_t(x|z)$. The **CFM loss**:

$$\mathcal{L}_{\text{CFM}}(\phi) = \mathbb{E}_{t\sim\mathcal{U}[0,1],\, z\sim p_{\text{data}},\, x\sim p_t(\cdot|z)}\left[\|v_\phi(x,t) - v_t(x|z)\|^2\right]$$

has the same gradient as the intractable marginal FM loss $\mathcal{L}_{\text{FM}}(\phi) = \mathbb{E}_{t,x\sim p_t}\|v_\phi(x,t) - v_t(x)\|^2$.

Proof

$\mathcal{L}_{\text{FM}} - \mathcal{L}_{\text{CFM}} = \mathbb{E}_{t,x}[\|v_t(x)\|^2 - \|\mathbb{E}[v_t(x|z)|x]\|^2]$ which is independent of ϕ. So $\nabla_\phi \mathcal{L}_{\text{FM}} = \nabla_\phi \mathcal{L}_{\text{CFM}}$. □

6.4.3. Gaussian / Linear Paths

The simplest choice: interpolate linearly between source and target.

$$x_t = (1-t)x_0 + t\,x_1, \quad x_0 \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}), \quad x_1 \sim p_{\text{data}}.$$

The conditional velocity is the constant direction from source to target:

$$v_t(x_t|x_0, x_1) = \frac{dx_t}{dt} = x_1 - x_0.$$

Gaussian CFM Training Loss

$$\mathcal{L}_{\text{CFM}}(\phi) = \mathbb{E}_{t,x_0\sim\mathcal{N}(\mathbf{0},\boldsymbol{I}),\, x_1\sim p_{\text{data}}}\left[\|v_\phi((1-t)x_0 + t\,x_1,\ t) - (x_1 - x_0)\|^2\right].$$

The network simply predicts the constant direction from noise to data.

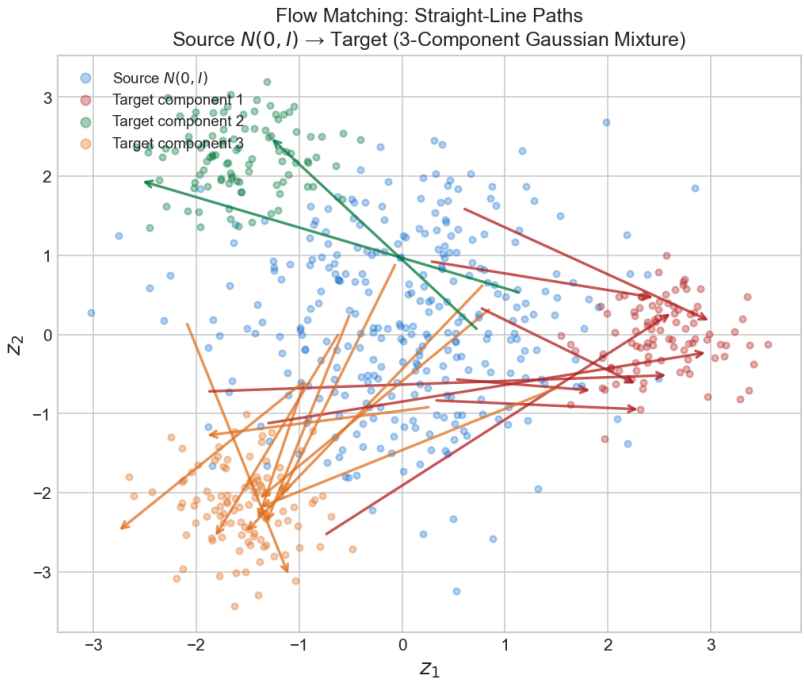


Figure 6.3: Gaussian flow matching: 20 straight-line sample paths from $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$ to a 3-component Gaussian mixture. Each path is a linear interpolation $(1-t)x_0 + tx_1$ with constant velocity $x_1 - x_0$. No ODE simulation is needed during training — only the endpoints.

6.4.4. The Unifying Framework: Score SDE and PF-ODE

Song et al. (2021) showed that the continuous-time DDPM forward process is described by a SDE:

$$dx = f(x,t)\,dt + g(t)\,dW_t,$$

and the corresponding reverse-time SDE (Anderson, 1982) is:

$$dx = [f(x,t) - g(t)^2 \nabla_x \log p_t(x)]\,dt + g(t)\,d\bar{W}_t.$$

The score $\nabla_x \log p_t(x)$ drives the reverse diffusion. The associated **Probability Flow ODE** (PF-ODE) — a deterministic process with the same marginals:

$$\frac{dx}{dt} = f(x,t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x)$$

is precisely a flow matching ODE with $v_t(x) = f(x,t) - \frac{1}{2} g(t)^2 s_\theta(x,t)$. The three frameworks — DDPM, Score SDE, Flow Matching — are three equivalent descriptions of the same transport problem.

Lecture 7

Causal Inference

Historical Note: A Brief History of Causal Inference

The mathematics of causality has two parallel traditions. The **potential outcomes** framework originates with Neyman (1923) in agricultural experiments and was developed by Rubin (1974) into a formal theory of causal effects, now standard in statistics and econometrics. The **graphical / structural** approach was developed by Wright (1921, path coefficients), Haavelmo (1944, structural equations), and most comprehensively by Pearl (1988–2009) with directed acyclic graphs and do-calculus. The two traditions are now known to be mathematically equivalent under the SUTVA assumption. A bridge was built by Richardson and Spirtes (2002), Robins (1986), and others.

7.1. Why Causation Cannot be Reduced to Correlation

Standard supervised ML minimizes $\mathbb{E}[\ell(f(X), Y)]$: the best prediction of Y given X under the *same* distribution. This is the right objective for prediction. But many scientific and policy questions require more:

- **Selection bias**: hospitals have higher mortality rates than non-hospitals observationally — patients self-select into hospitals because they are sick. Sending people to hospitals does not cause mortality; the confounded observation $P(\text{death}|\text{hospital})$ is misleading.
- **Distribution shift**: a model trained to predict unemployment from income learns $P(\text{unemployed}|X)$. If we implement a policy that changes X, the learned f may give completely wrong predictions.
- **Policy evaluation**: does increasing the minimum wage reduce employment? The observational correlation between minimum wage and employment conflates the causal effect with reverse causation (employment drives local political demand for wage increases).

Causal inference formalizes the distinction between $P(Y|X = x)$ (observational: "what do we see when $X = x$?") and $P(Y|\text{do}(X = x))$ (interventional: "what happens when we externally *set* $X = x$?").

7.2. Pearl's Ladder of Causation

Three Rungs of Causation (Pearl, 2009)

Rung 1. Association — $P(Y|X = x)$. Pure observational statistics. Answerable from data alone. Example: "Patients taking drug X have better outcomes."

Rung 2. Intervention — $P(Y|\text{do}(X = x))$. Fix X externally, breaking its dependence on confounders. Requires structural assumptions beyond data. Example: "What happens if we prescribe drug X to this population?"

Rung 3. Counterfactual — $P(Y_{x'}|X = x, Y = y)$. Given that individual i took $X = x$ and had outcome $Y = y$, what would have happened had they taken $X = x'$? Requires a structural causal model (SCM). Example: "If patient i had not taken drug X, would they have survived?"

Key result: No amount of data alone can climb the ladder. Each rung requires additional structural assumptions.

7.3. The Potential Outcomes Framework (Rubin, 1974)

Definition: Potential Outcomes

For each unit $i = 1, \ldots, n$: binary treatment $T_i \in \{0, 1\}$, covariates $X_i \in \mathbb{R}^p$, and potential outcomes:

- $Y_i(1)$: outcome unit i would have if treated.
- $Y_i(0)$: outcome unit i would have if untreated.
- **Individual treatment effect (ITE)**: $\tau_i = Y_i(1) - Y_i(0)$.

We observe only: $Y_i^{\text{obs}} = T_i Y_i(1) + (1 - T_i)Y_i(0)$.

The Fundamental Problem of Causal Inference (Holland, 1986)

$Y_i(1 - T_i)$ is the *counterfactual* outcome — it is never observed. We cannot directly compute τ_i for any individual. This is why causal inference requires averaging and identifying assumptions.

SUTVA (Stable Unit Treatment Value Assumption): (i) no interference — $Y_i(T_i)$ does not depend on T_j for $j \neq i$; (ii) no hidden versions of treatment — $T_i = 1$ means the same treatment for all units. Without SUTVA, $Y_i(T_i)$ is not even well-defined.

7.3.1. Estimands

Quantity	Definition	Identified by
ATE	$\mathbb{E}[Y(1) - Y(0)]$	Ignorability + overlap
ATT	$\mathbb{E}[Y(1) - Y(0)\|T = 1]$	Ignorability
ATC	$\mathbb{E}[Y(1) - Y(0)\|T = 0]$	Ignorability
CATE	$\tau(x) = \mathbb{E}[Y(1) - Y(0)\|X = x]$	Ignorability + overlap
LATE	$\mathbb{E}[Y(1) - Y(0)\|\text{complier}]$	IV assumptions

7.3.2. Identification under Ignorability

Identification Theorem

Assume:
(i) **Ignorability / Unconfoundedness**: $\{Y(0), Y(1)\} \perp\!\!\!\perp T \mid X$.
(ii) **Positivity / Overlap**: $0 < P(T = 1|X = x) < 1$ for almost all x.
Then $\mathbb{E}[Y(t)|X = x] = \mathbb{E}[Y^{\text{obs}}|T = t, X = x]$, and:

$$\text{ATE} = \mathbb{E}_X[\mathbb{E}[Y|T = 1, X] - \mathbb{E}[Y|T = 0, X]]\,.$$

Proof

Under ignorability:

$$\begin{aligned}\mathbb{E}[Y(t)|X = x] &= \mathbb{E}[Y(t)|T = t, X = x] \quad \text{(ignorability: } Y(t) \perp\!\!\!\perp T|X\text{)}\\ &= \mathbb{E}[Y^{\text{obs}}|T = t, X = x] \quad \text{(consistency: } Y^{\text{obs}} = Y(T) \text{ and } T = t\text{)}.\end{aligned}$$

Integrating over X with appropriate weighting gives the ATE formula. □

Naive estimator bias.

$$\mathbb{E}[Y|T = 1] - \mathbb{E}[Y|T = 0] = \underbrace{\mathbb{E}[Y(1) - Y(0)|T = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y(0)|T = 1] - \mathbb{E}[Y(0)|T = 0]}_{\text{selection bias}}\,.$$

Selection bias is zero only when treatment is assigned randomly. Ignorability (conditioning on X) kills the selection bias.

7.4. Structural Causal Models and DAGs

Definition: Structural Causal Model (SCM)

An SCM $\mathcal{M}$ consists of:
- Endogenous variables $V = (X_1, \ldots, X_p)$ with exogenous noise $U = (U_1, \ldots, U_p)$.
- Structural equations $X_i := f_i(\text{Pa}(X_i), U_i)$, where $\text{Pa}(X_i)$ are the direct *causal parents* of X_i in the graph.
- The associated DAG G has edge $X_j \to X_i$ iff $X_j \in \text{Pa}(X_i)$.

Markov factorization: $P(X_1, \ldots, X_p) = \prod_{i=1}^{p} P(X_i|\text{Pa}(X_i))$.

7.4.1. The Three Elementary Path Motifs

- **Chain** $X \to Z \to Y$: $X \not\perp\!\!\!\perp Y$ marginally; $X \perp\!\!\!\perp Y|Z$. Conditioning on Z *blocks* the path. Z is a *mediator*.

- **Fork** $X \leftarrow Z \to Y$: $X \not\perp\!\!\!\perp Y$ marginally; $X \perp\!\!\!\perp Y|Z$. Z is a *common cause* (confounder). Conditioning on Z removes spurious association.

- **Collider** $X \to Z \leftarrow Y$: $X \perp\!\!\!\perp Y$ marginally; $X \not\perp\!\!\!\perp Y|Z$. Z is a *collider*. Conditioning on Z *opens* the path, creating spurious dependence (*collider bias* or *Berkson's paradox*).

Example: Berkson's Paradox / Collider Bias

Among patients hospitalized (conditioning on admission $Z = 1$), there is a negative correlation between disease severity X and accident severity Y, even if they are independent in the population. The reason: to be hospitalized, a patient needs either severe disease OR a serious accident. Knowing one cause (X is mild) makes the other (Y is severe) more likely — a collider bias. This explains counterintuitive associations in hospital-based studies.

d-Separation (Pearl, 1988)

A path π between X and Y is **blocked** by a set Z if:

(i) π contains a chain $A \to M \to B$ or fork $A \leftarrow M \to B$ with $M \in Z$; or

(ii) π contains a collider $A \to M \leftarrow B$ with $M \notin Z$ and no descendant of M in Z.

X and Y are **d-separated** by Z (written $X \perp_G Y|Z$) if every path is blocked.
Under **faithfulness** (no cancellations in P): $X \perp_G Y|Z \iff X \perp\!\!\!\perp_P Y|Z$.

Backdoor Criterion (Pearl, 1993)

A set Z satisfies the **backdoor criterion** relative to (X, Y) if:

(i) No node in Z is a descendant of X.

(ii) Z blocks every path from X to Y with an arrow into X ("backdoor paths").

If Z satisfies the backdoor criterion:

$$P(Y|\mathrm{do}(X = x)) = \sum_z P(Y|X = x, Z = z)\, P(Z = z) \quad \text{(backdoor adjustment)}.$$

Proof sketch

The do-operator on X removes all incoming arrows to X, creating the mutilated graph $G_{\bar{X}}$. In $G_{\bar{X}}$, there are no backdoor paths. By the truncated factorization:

$$P(Y|\mathrm{do}(X = x)) = \sum_z P(Y|X = x, Z = z)\, P(Z = z|\mathrm{do}(X = x)).$$

Since Z has no descendant of X in it, $P(Z|\mathrm{do}(X = x)) = P(Z)$ in $G_{\bar{X}}$ (removing arrows to X does not affect Z's distribution). □

7.5. Instrumental Variables

7.5.1. When Backdoor Fails: Hidden Confounders

Suppose there is an unobserved confounder U with $T \leftarrow U \to Y$. No subset Z of observed variables blocks all backdoor paths through U. The ATE is not identified from observational data without additional structure.

Instrumental Variables (IV) exploit a third variable that affects treatment but not the outcome directly.

Definition: Valid Instrument

A variable Z is a valid instrument for the effect of T on Y if:

(i) **Relevance**: $Z \not\perp\!\!\!\perp T$ (the instrument moves treatment; $\mathrm{Cov}(Z, T) \neq 0$).
(ii) **Exclusion**: Z affects Y only through T (no direct path $Z \to Y$).
(iii) **Independence**: $Z \perp\!\!\!\perp U$ (the instrument is as good as randomly assigned).

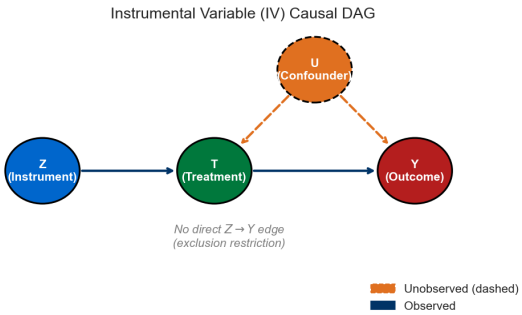


Figure 7.1: Instrumental variable DAG: Z (instrument) affects treatment T, which affects outcome Y. The unobserved confounder U (dashed) affects both T and Y. Crucially, Z has no direct path to Y and is independent of U.

Example: Draft Lottery as an Instrument for Military Service

Angrist (1990) used the Vietnam-era draft lottery as an instrument for military service. Z = draft lottery number (randomly assigned), T = military service, Y = future earnings. The lottery is relevant (high lottery numbers increase service), satisfies exclusion (lottery number does not directly affect earnings), and is independent of confounders (it was randomized). The IV estimate found that military service reduced civilian earnings.

7.5.2. The Wald Estimator and LATE

Wald Estimator (binary $Z \in \{0, 1\}$)

$$\hat{\beta}_{\mathrm{IV}} = \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[T|Z=1] - \mathbb{E}[T|Z=0]} = \frac{\mathrm{Cov}(Y, Z)}{\mathrm{Cov}(T, Z)}.$$

LATE Theorem (Angrist, Imbens & Rubin, 1996)

Define compliance types (binary Z, binary T):

Type	$T(Z=0)$	$T(Z=1)$	Proportion	Affected by Z?
Complier	0	1	π_c	Yes
Always-taker	1	1	π_a	No
Never-taker	0	0	π_n	No
Defier	1	0	π_d	Yes (reversed)

Under relevance, exclusion, independence, and **monotonicity** ($T_i(Z=1) \geq T_i(Z =$

0) for all i, ruling out defiers):

$$\hat{\beta}_{\text{IV}} \xrightarrow{p} \text{LATE} = \mathbb{E}[Y(1) - Y(0)|\text{complier}].$$

Proof

Under exclusion, $Y_i(Z) = Y_i(T_i(Z))$:

$$\begin{aligned}\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0] &= \mathbb{E}[Y(T(1)) - Y(T(0))] \\ &= \mathbb{E}[Y(1) - Y(0)|T(1)=1, T(0)=0] \cdot P(\text{complier}) \\ &\quad + \mathbb{E}[Y(1) - Y(0)|T(1)=0, T(0)=1] \cdot P(\text{defier}) \\ &= \text{LATE} \cdot \pi_c - \mathbb{E}[Y(1) - Y(0)|\text{defier}] \cdot \pi_d.\end{aligned}$$

Under monotonicity: $\pi_d = 0$, so $\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0] = \text{LATE} \cdot \pi_c$. The denominator $\mathbb{E}[T|Z=1] - \mathbb{E}[T|Z=0] = \pi_c - \pi_d = \pi_c$. Dividing gives LATE. □

The LATE limitation. IV identifies the causal effect only for *compliers* — the subpopulation whose treatment is actually moved by the instrument. This subpopulation may not be representative of the full population, limiting external validity. Different instruments identify effects for different complier subpopulations.

7.5.3. Manski Partial Identification Bounds

When IV assumptions fail (or no instrument is available), Manski (1990, 1995) showed that the ATE can be bounded — without point identification — using only monotone assumptions.

Manski Bounds (under no assumptions beyond SUTVA)

If $Y \in [y_{\min}, y_{\max}]$:

$$\begin{aligned}\text{ATE} &\geq \mathbb{E}[Y|T=1]P(T=1) + y_{\min}P(T=0) - \mathbb{E}[Y|T=0]P(T=0) - y_{\max}P(T=1) \\ \text{ATE} &\leq \mathbb{E}[Y|T=1]P(T=1) + y_{\max}P(T=0) - \mathbb{E}[Y|T=0]P(T=0) - y_{\min}P(T=1).\end{aligned}$$

The width of the interval is $(y_{\max} - y_{\min}) \cdot P(T=0)P(T=1) \cdot 2$, which is wide unless treatment is nearly universal or nearly absent.

Additional restrictions (monotone treatment response, monotone treatment selection) tighten the bounds. This approach is *honest*: it reports exactly what the data alone can tell us.

7.6. Causal Discovery: The PC Algorithm

All previous methods assumed the causal graph is known. **Causal discovery** recovers the graph from observational data.

Definition: PC Algorithm (Spirtes, Glymour & Scheines, 1993)

Step 1. **Skeleton recovery**: start with a complete undirected graph. For each pair (X_i, X_j) and each set $S \subseteq V \setminus \{i, j\}$: run a conditional independence test for $X_i \perp\!\!\!\perp X_j | S$. If independence is detected, remove edge $X_i - X_j$ and record the *separating set* $\mathrm{Sep}(i, j) = S$.

Step 2. **V-structure orientation**: for each unshielded triple $X_i - X_k - X_j$ with X_i and X_j non-adjacent, check if $X_k \in \mathrm{Sep}(i, j)$. If not, orient as $X_i \to X_k \leftarrow X_j$ (collider).

Step 3. **Meek orientation rules**: apply deterministic rules to orient remaining edges without creating new v-structures or directed cycles. Repeat until no more rules apply.

Consistency of PC (Spirtes et al., 1993; Meek, 1995)

If the conditional independence tests are consistent and the faithfulness assumption holds, the PC algorithm returns the **completed partially directed acyclic graph (CPDAG)** that represents the Markov equivalence class of the true DAG, with probability tending to 1 as $n \to \infty$.

Markov equivalence class. Many DAGs can encode the same set of conditional independencies (same skeleton and v-structures). The PC algorithm identifies the *equivalence class* — it can orient edges that have the same direction in all equivalent DAGs, but not edges that could be reversed without changing the encoded independencies.

Lecture 8

Double Machine Learning

Historical Note: From Econometrics to Double Machine Learning

The problem of estimating a low-dimensional parameter of interest while controlling for high-dimensional nuisances has a long history. Robinson (1988) introduced the partially linear model and showed that $\sqrt{n}$-consistent inference on θ_0 is possible even with nonparametric nuisance estimation, provided the nuisance rates are fast enough. Newey (1994) developed the theory of semiparametric efficiency. The modern formulation with explicit orthogonalization and cross-fitting is due to Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey and Robins (2018), who combined Robinson's decomposition with Neyman's (1959) orthogonality idea and the sample-splitting of Schick (1986). The resulting estimator — Double Machine Learning — enables rigorous inference in high-dimensional settings using any machine learning method satisfying mild rate conditions.

8.1. The Problem: High-Dimensional Controls with Valid Inference

Example: 401(k) Eligibility and Savings (Chernozhukov et al., 2018)

Does 401(k) plan eligibility increase net financial assets?

- Y: net financial assets (savings, continuous).
- D: 401(k) eligibility (binary treatment).
- X: income, age, education, family size, marital status, IRA participation ($p > 100$ controls).

Eligible workers differ systematically: higher-income, older, better-educated workers are more often eligible. We must flexibly control for X.

The fundamental tension. Machine learning handles high-dimensional X flexibly — but introduces regularization bias of order $O_{\mathbb{P}}(n^{-1/4})$, which is much larger than the $O_{\mathbb{P}}(n^{-1/2})$ variance. Confidence intervals built from a naive ML plug-in have coverage $\to 0$.

8.2. The Partially Linear Model

Definition: Partially Linear Model (PLM, Robinson 1988)

$$Y_i = \theta_0 D_i + g_0(X_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i | D_i, X_i] = 0,$$

where Y_i is outcome, $D_i \in \mathbb{R}$ is a scalar treatment, $X_i \in \mathbb{R}^p$ are controls, $g_0 : \mathbb{R}^p \to \mathbb{R}$ is an unknown function, and $\theta_0 \in \mathbb{R}$ is the parameter of interest.

Key assumptions:

(i) **Unconfoundedness**: $\varepsilon_i \perp\!\!\!\perp D_i | X_i$. Conditional on X, treatment is as good as random.

(ii) **Partial linearity**: the effect of D on Y is linear (with slope θ_0), but g_0 is unrestricted.

(iii) **High-dimensionality**: p may exceed n.

Goal: estimate θ_0 and construct a 95% CI with asymptotically correct coverage.

8.3. Why Naive ML Fails

Short regression (omit X): $\hat{\theta}_{\text{short}} = \theta_0 + \text{Cov}(D, g_0(X))/\text{Var}(D)$ — confounding bias; inconsistent.

Direct LASSO: regress Y on (D, X) with LASSO penalty on $\beta = (\theta, g)$. The LASSO shrinks θ toward zero along with the other coefficients — the estimate of θ_0 is biased of order $O_{\mathbb{P}}(n^{-1/4})$.

Naive plug-in: estimate g_0 by ML (LASSO/RF), denote $\hat{g}$, then run OLS of $Y - \hat{g}(X)$ on D:

$$\hat{\theta}^{\text{naive}} = \left(\frac{1}{n}\sum_i D_i^2\right)^{-1} \frac{1}{n}\sum_i D_i(Y_i - \hat{g}(X_i)).$$

Bias of Naive Plug-in

Write $\sqrt{n}(\hat{\theta}^{\text{naive}} - \theta_0) = a + b$ where:

$$a = \left(\frac{1}{n}\sum_i D_i^2\right)^{-1} \frac{1}{\sqrt{n}}\sum_i D_i \varepsilon_i \xrightarrow{d} N(0, \Sigma),$$

$$b = \left(\frac{1}{n}\sum_i D_i^2\right)^{-1} \frac{1}{\sqrt{n}}\sum_i D_i (g_0 - \hat{g})(X_i).$$

By Cauchy-Schwarz: $|b| \leq O_{\mathbb{P}}(1) \cdot \sqrt{n} \cdot \|g_0 - \hat{g}\|_2 = O_{\mathbb{P}}(1) \cdot n^{1/2} \cdot O_{\mathbb{P}}(n^{-1/4}) = O_{\mathbb{P}}(n^{1/4}) \to \infty$.

The bias b grows with n — confidence intervals have zero asymptotic coverage.

8.4. Robinson's Decomposition

8.4.1. Frisch-Waugh-Lovell (FWL) Theorem

In the linear model $Y = D\theta + X\beta + e$, the FWL theorem states:

$$\hat{\theta}_{\text{OLS}} = \frac{\tilde{D}^\top \tilde{Y}}{\tilde{D}^\top \tilde{D}}, \qquad \tilde{D} = M_X D, \quad \tilde{Y} = M_X Y,$$

where $M_X = I - X(X^\top X)^{-1}X^\top$ is the residual-maker. Only the variation in D *orthogonal to* X drives the estimate.

8.4.2. Nonparametric Extension: Robinson's Equation

In the PLM, define:

$$m_0(x) = \mathbb{E}[D|X = x], \qquad \ell_0(x) = \mathbb{E}[Y|X = x].$$

Taking conditional expectations in the PLM: $\ell_0(X_i) = \theta_0 m_0(X_i) + g_0(X_i)$. Subtracting:

$$\underbrace{Y_i - \ell_0(X_i)}_{\widetilde{Y}_i} = \theta_0 \underbrace{(D_i - m_0(X_i))}_{V_i} + \varepsilon_i.$$

Robinson's Equation

$\widetilde{Y}_i = \theta_0 V_i + \varepsilon_i$. The nuisance g_0 has been completely eliminated.
If m_0, ℓ_0 were known: $\hat{\theta}^{\text{oracle}} = \frac{\mathbb{E}_n[V_i \widetilde{Y}_i]}{\mathbb{E}_n[V_i^2]}$ is $\sqrt{n}$-consistent and asymptotically normal: $\sqrt{n}(\hat{\theta}^{\text{oracle}} - \theta_0) \xrightarrow{d} N(0, \Sigma)$.

Proof

Oracle unbiasedness: $\mathbb{E}_n[V_i \widetilde{Y}_i] = \theta_0 \mathbb{E}_n[V_i^2] + \mathbb{E}_n[V_i \varepsilon_i]$. Since $V_i = D_i - m_0(X_i)$ satisfies $\mathbb{E}[V_i|\varepsilon_i, X_i] = 0$, the term $\mathbb{E}_n[V_i \varepsilon_i] = O_{\mathbb{P}}(n^{-1/2})$. By CLT: $\sqrt{n}\mathbb{E}_n[V_i \varepsilon_i] \xrightarrow{d} N(0, \mathbb{E}[V^2 \varepsilon^2])$. Delta method gives the result. $\square$

8.5. The DML Solution

DML achieves $\sqrt{n}$-consistent inference via two key innovations: **Neyman orthogonality** and **cross-fitting**.

8.5.1. From $a + b$ to $a^* + b^* + c^*$

The DML estimator uses estimated residuals $\hat{V}_i = D_i - \hat{m}(X_i)$:

$$\hat{\theta}^{\text{DML}} = \left(\frac{1}{n} \sum_i \hat{V}_i D_i \right)^{-1} \frac{1}{n} \sum_i \hat{V}_i (Y_i - \hat{\ell}(X_i)).$$

Error Decomposition of DML: $a^* + b^* + c^*$

$\sqrt{n}(\hat{\theta}^{\text{DML}} - \theta_0) = a^* + b^* + c^*$, where:

$$a^* = (\mathbb{E}[V^2])^{-1} \frac{1}{\sqrt{n}} \sum_i V_i \varepsilon_i \xrightarrow{d} N(0, \Sigma) \quad \text{(signal)},$$

$$b^* = (\mathbb{E}[V^2])^{-1} \frac{1}{\sqrt{n}} \sum_i \underbrace{(\hat{m} - m_0)(X_i)}_{\text{error in } \hat{m}} \cdot \underbrace{(\hat{\ell} - \ell_0)(X_i)}_{\text{error in } \hat{\ell}} \quad \text{(product of ML errors)},$$

$$c^* = \text{cross-terms involving } \hat{m} - m_0 \text{ and } \varepsilon_i \quad \text{(controlled by cross-fitting)}.$$

The magic of DML: b^* is a *product* of two ML estimation errors. By Cauchy-Schwarz:

$$|b^*| \leq O_{\mathbb{P}}(1) \cdot \sqrt{n} \cdot \|\hat{m} - m_0\|_2 \cdot \left\|\hat{\ell} - \ell_0\right\|_2 = O_{\mathbb{P}}(n^{1/2}) \cdot O_{\mathbb{P}}(n^{-1/4}) \cdot O_{\mathbb{P}}(n^{-1/4}) = o_{\mathbb{P}}(1).$$

8.5.2. Ingredient 1: Neyman Orthogonality

Definition: Neyman Orthogonality (Neyman, 1959)

A score function $\psi(W; \theta, \eta)$ is **Neyman orthogonal** at (θ_0, η_0) if:

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta_0)][\delta\eta] = 0 \quad \forall \text{ directions } \delta\eta.$$

Consequence: bias from $\hat{\eta}$ enters $\hat{\theta}$ only at second order: $\text{Bias}(\hat{\theta}) = O_{\mathbb{P}}(\|\hat{\eta} - \eta_0\|^2)$ instead of $O_{\mathbb{P}}(\|\hat{\eta} - \eta_0\|)$.

The naive estimating equation uses score $\psi^{\text{naive}}(W; \theta) = D \cdot (Y - D\theta - \hat{g}(X))$. Errors in $\hat{g}$ enter linearly (first order) into the score.

The DML score uses $V_i = D_i - m_0(X_i)$:

$$\psi^{\text{DML}} = (\widetilde{Y}_i - \theta V_i) \cdot V_i = (Y_i - D_i\theta - \ell_0(X_i) + \theta m_0(X_i)) \cdot (D_i - m_0(X_i)).$$

Orthogonality of PLM Score

ψ^{DML} is Neyman orthogonal: errors in both $\hat{\ell}$ and $\hat{m}$ vanish at first order.

Proof

Perturb ℓ (outcome nuisance): $\partial_\ell \mathbb{E}[\psi][\delta\ell] = -\mathbb{E}[\delta\ell(X) \cdot V] = -\mathbb{E}[\delta\ell(X) \cdot \mathbb{E}[V|X]] = 0$, since $\mathbb{E}[V|X] = \mathbb{E}[D - m_0(X)|X] = 0$ by construction.
Perturb m (propensity nuisance): $\partial_m \mathbb{E}[\psi][\delta m] = -\theta \mathbb{E}[\delta m(X) \cdot V] + \mathbb{E}[\varepsilon \cdot \delta m(X)]$. First term: $\mathbb{E}[\delta m(X) \cdot V] = 0$ as above. Second term: $\mathbb{E}[\varepsilon \cdot \delta m(X)] = \mathbb{E}[\delta m(X) \cdot \mathbb{E}[\varepsilon|X]] = 0$ since $\mathbb{E}[\varepsilon|X] = 0$ by the model assumption. □

8.5.3. Ingredient 2: Cross-Fitting

Definition: K-Fold Cross-Fitting (Schick, 1986; Chernozhukov et al., 2018)

1. Split the sample into K folds $I_1, \ldots, I_K$ of size n/K.
2. For each fold k: train $\hat{m}_{-k}$ and $\hat{\ell}_{-k}$ on $I_k^c = \bigcup_{j \neq k} I_j$ (all data except fold k).
3. Predict out-of-fold residuals for $i \in I_k$: $\hat{V}_i = D_i - \hat{m}_{-k}(X_i)$, $\hat{\tilde{Y}}_i = Y_i - \hat{\ell}_{-k}(X_i)$.
4. Pool all folds: $\hat{\theta}^{\text{DML2}} = \left(\sum_i \hat{V}_i D_i\right)^{-1} \sum_i \hat{V}_i (Y_i - \hat{\ell}_{-k(i)}(X_i))$.

Why cross-fitting kills c^*. Without cross-fitting, $\hat{m}(X_i)$ is trained on the same ε_i it is evaluated against. The empirical process $\frac{1}{\sqrt{n}} \sum_i (\hat{m}(X_i) - m_0(X_i))\varepsilon_i$ involves a correlation between the estimated function (dependent on the data) and the noise. This does not vanish even for well-regularized ML methods, unless the hypothesis class is a Donsker class (which modern ML methods are not).

With cross-fitting: $\hat{m}_{-k}(X_i)$ was trained without seeing (X_i, D_i, Y_i), so $\hat{m}_{-k}(X_i) - m_0(X_i)$ is independent of ε_i conditional on the training fold. The expected cross-term is exactly zero: $\mathbb{E}[(\hat{m}_{-k}(X_i) - m_0(X_i))\varepsilon_i | S_{-k}] = \mathbb{E}[\mathbb{E}[\varepsilon_i | X_i] \cdot (\hat{m}_{-k}(X_i) - m_0(X_i)) | S_{-k}] = 0$.

Main DML Theorem (Chernozhukov et al., 2018)

Under the PLM, unconfoundedness, and the rate condition:

$$\|\hat{m} - m_0\|_2 \cdot \left\|\hat{\ell} - \ell_0\right\|_2 = o_{\mathbb{P}}(n^{-1/2}),$$

the DML2 estimator satisfies:

$$\sqrt{n}(\hat{\theta}^{\text{DML}} - \theta_0) \xrightarrow{d} N(0, \Sigma),$$

where $\Sigma = (\mathbb{E}[V^2])^{-1}\mathbb{E}[V^2\varepsilon^2](\mathbb{E}[V^2])^{-1}$ is the heteroskedasticity-robust (sandwich) variance. A valid 95% CI: $\hat{\theta} \pm 1.96\hat{\sigma}/\sqrt{n}$.

Proof sketch

After cross-fitting, the error decomposes as $\sqrt{n}(\hat{\theta} - \theta_0) = a^* + b^* + c^*$. We need to show $b^*, c^* = o_{\mathbb{P}}(1)$.
Term b^*: by Cauchy-Schwarz and the rate condition $\|\hat{m} - m_0\|_2 \left\|\hat{\ell} - \ell_0\right\|_2 = o_{\mathbb{P}}(n^{-1/2})$:

$$|b^*| \leq \sqrt{n} \cdot \|\hat{m} - m_0\|_2 \left\|\hat{\ell} - \ell_0\right\|_2 = \sqrt{n} \cdot o_{\mathbb{P}}(n^{-1/2}) = o_{\mathbb{P}}(1).$$

Term c^*: cross-fitting ensures $\mathbb{E}[c^* | S_{-k}] = 0$. By a Bernstein-type inequality for empirical processes with the cross-fitting independence structure: $|c^*| \leq O_{\mathbb{P}}(\|\hat{\eta} - \eta_0\|^2 \sqrt{n}) = o_{\mathbb{P}}(1)$ under the rate condition.
Term a^*: CLT for $\frac{1}{\sqrt{n}} \sum_i V_i \varepsilon_i$ using $\mathbb{E}[V_i \varepsilon_i] = 0$ (unconfoundedness) and $\mathbb{E}[V_i^2 \varepsilon_i^2] < \infty$.
Combining: $\sqrt{n}(\hat{\theta} - \theta_0) = a^* + o_{\mathbb{P}}(1) \xrightarrow{d} N(0, \Sigma)$. □

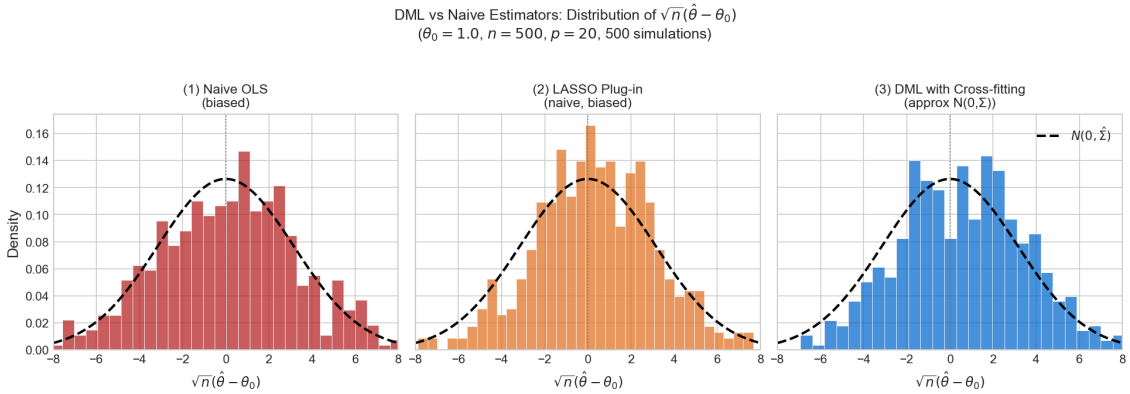


Figure 8.1: Sampling distributions of $\sqrt{n}(\hat{\theta} - \theta_0)$ from 500 simulations ($n = 500$, $p = 20$). Left: naive OLS plug-in — severely biased, non-normal. Center: LASSO plug-in — biased, shifted from zero. Right: DML with cross-fitting — centered on zero, well-approximated by $N(0, \Sigma)$. Only DML provides valid inference.

8.6. When Are Rate Conditions Achievable?

The DML rate condition is $\|\hat{m} - m_0\| \left\|\hat{\ell} - \ell_0\right\| = o_{\mathbb{P}}(n^{-1/2})$. If each nuisance achieves $O_{\mathbb{P}}(n^{-1/4})$, the product is $o_{\mathbb{P}}(n^{-1/2})$.

Method	Assumption	Rate per nuisance	DML condition
LASSO	$s \log p = o(\sqrt{n})$	$\sqrt{s \log p / n}$	Satisfied
RF (Hölder β)	$\beta > d/2$	$n^{-\beta/(2\beta+d)}$	Satisfied
RF (additive)	Scornet et al.	$n^{-2/(d+2)}$ (Lipschitz)	Satisfied for $d \leq 2$
Neural Net	Barron class	$\sim n^{-1/3}$ (sparse)	Satisfied

LASSO case. With sparsity s in both m_0 and g_0, and $\lambda \asymp \sigma\sqrt{\log p / n}$:

$$\|\hat{m} - m_0\|_2 \left\|\hat{\ell} - \ell_0\right\|_2 \leq \frac{s \log p}{n} = o_{\mathbb{P}}(n^{-1/2}) \iff s^2 (\log p)^2 = o(n^{3/2}).$$

This is satisfied when $s = O(n^{3/8}/(\log p)^{3/4})$ — e.g., $n = 1000$, $p = 10000$, $s \leq 10$.

8.7. The Interactive Model and Doubly Robust Estimation

The PLM assumes Y depends linearly on D, which is restrictive when effects are heterogeneous. The **interactive (interactive regression) model** allows:

$$Y_i = g_0(D_i, X_i) + \varepsilon_i, \qquad D_i \in \{0, 1\}, \qquad \mathbb{E}[\varepsilon_i | D_i, X_i] = 0.$$

Target: ATE $= \theta_0 = \mathbb{E}[g_0(1, X) - g_0(0, X)]$.

Definition: Doubly Robust (DR / AIPW) Score

$$\psi^{\mathrm{DR}} = g_0(1, X) - g_0(0, X) + \frac{D}{\pi(X)}[Y - g_0(1, X)] - \frac{1 - D}{1 - \pi(X)}[Y - g_0(0, X)],$$

where $\pi(x) = P(D = 1|X = x)$ is the propensity score.

Double Robustness

$\mathbb{E}[\psi^{\mathrm{DR}}] = \mathrm{ATE}$ provided either g_0 or π is correctly specified (but not necessarily both).

Proof

If g_0 is correct: $\mathbb{E}[\psi^{\mathrm{DR}}] = \mathbb{E}[g_0(1, X) - g_0(0, X)] + \mathbb{E}\left[\frac{D\varepsilon}{\pi(X)}\right] - \mathbb{E}\left[\frac{(1-D)\varepsilon}{1-\pi(X)}\right]$. The last two terms: $\mathbb{E}\left[\frac{D\varepsilon}{\pi(X)}\right] = \mathbb{E}\left[\frac{\mathbb{E}[D|X]\varepsilon}{\pi(X)}\right] = \mathbb{E}[\varepsilon] = 0$ by the tower property and the model. Similarly for the other term.
If π is correct: $\mathbb{E}[\psi^{\mathrm{DR}}] = \mathbb{E}\left[\frac{D}{\pi(X)}Y\right] - \mathbb{E}\left[\frac{1-D}{1-\pi(X)}Y\right]$ (after algebra). By Horvitz-Thompson, this equals $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \mathrm{ATE}$. □

Neyman orthogonality of DR score. Errors in $\hat{g}$ and $\hat{\pi}$ enter only at second order. Applying cross-fitting yields the DML estimator for ATE with valid $\sqrt{n}$-inference, under the rate condition $\|\hat{g} - g_0\| \, \|\hat{\pi} - \pi\| = o_{\mathbb{P}}(n^{-1/2})$.

8.8. Instrumental Variables in the DML Framework

Note: this section extends beyond the lecture content.

When unconfoundedness fails but a valid instrument Z is available, we work with the **partially linear IV model**:

$$Y_i = \theta_0 D_i + g_0(X_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i | Z_i, X_i] = 0,$$

where ε_i may be correlated with D_i (endogeneity), but Z_i satisfies relevance, exclusion, and independence.

8.8.1. DML-IV Estimator

Define residuals:

$$\tilde{D}_i = D_i - r_0(Z_i, X_i) \quad \text{where } r_0(z, x) = \mathbb{E}[D|Z = z, X = x],$$

$$\tilde{Z}_i = Z_i - q_0(X_i) \quad \text{where } q_0(x) = \mathbb{E}[Z|X = x].$$

The IV moment condition $\mathbb{E}[\tilde{Z}_i(Y_i - \theta_0 D_i - g_0(X_i))] = 0$ motivates:

$$\hat{\theta}^{\mathrm{DML-IV}} = \frac{\mathbb{E}_n[\hat{\tilde{Z}}_i \hat{\tilde{Y}}_i]}{\mathbb{E}_n[\hat{\tilde{Z}}_i \hat{\tilde{D}}_i]},$$

where all nuisances $(\hat{g}, \hat{r}, \hat{q})$ are estimated via ML with cross-fitting.

DML-IV Asymptotic Normality

Under IV assumptions, unconfoundedness of Z, and rate conditions $\|\hat{\eta}_j - \eta_j\| = o_{\mathbb{P}}(n^{-1/4})$ for each nuisance j:

$$\sqrt{n}(\hat{\theta}^{\mathrm{DML-IV}} - \theta_0) \xrightarrow{d} N\left(0,\ \frac{\mathbb{E}[\tilde{Z}^2\varepsilon^2]}{(\mathbb{E}[\tilde{Z}\tilde{D}])^2}\right).$$

Special cases.

- No controls (X absent), binary Z: reduces to Wald estimator $\hat{\theta} = \mathrm{Cov}(Y, Z)/\mathrm{Cov}(D, Z)$.
- Linear controls (g_0 linear): reduces to 2SLS.
- Nonparametric controls: the first-ever valid $\sqrt{n}$-inference for nonparametric IV.

8.8.2. Weak Instruments

A critical concern is **instrument strength**. When $\mathbb{E}[\tilde{Z}\tilde{D}] \approx 0$ (weak first stage), the denominator of $\hat{\theta}^{\mathrm{DML-IV}}$ is near zero, inflating variance catastrophically. Standard Wald-based tests have size distortions; use:

- **Anderson-Rubin test**: inverts the test $H_0 : \theta = \theta_0$ using the first-stage reduced form. Exact size under Gaussian errors, robust to weak instruments.
- **First-stage F-test**: a rule of thumb $F > 10$ indicates adequate strength (Stock & Yogo, 2005), though modern theory suggests higher thresholds.
- **Conditional Likelihood Ratio (CLR) test**: optimal power among weak-instrument-robust tests.

Bibliography

Statistical Learning Theory

- Bartlett, P. L. & Mendelson, S. (2002). "Rademacher and Gaussian complexities: risk bounds and structural results." *JMLR*, 3, 463–482.
- Shalev-Shwartz, S. & Ben-David, S. (2014). *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press.
- Wainwright, M. J. (2019). *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*. Cambridge University Press.
- Bartlett, P., Montanari, A. & Rakhlin, A. (2021). "Deep learning: a statistical viewpoint." *Acta Numerica*.
- Vapnik, V. & Chervonenkis, A. (1971). "On the uniform convergence of relative frequencies of events to their probabilities." *Theory of Probability and its Applications*, 16(2), 264–280.
- Valiant, L. (1984). "A theory of the learnable." *Communications of the ACM*, 27(11), 1134–1142.

Regularisation

- Hastie, T., Tibshirani, R. & Friedman, J. (2009). *The Elements of Statistical Learning*, 2nd ed. Springer.
- Tibshirani, R. (1996). "Regression shrinkage and selection via the lasso." *JRSS-B*, 58(1), 267–288.
- Bickel, P., Ritov, Y. & Tsybakov, A. (2009). "Simultaneous analysis of Lasso and Dantzig selector." *Annals of Statistics*, 37(4), 1705–1732.
- Efron, B., Hastie, T., Johnstone, I. & Tibshirani, R. (2004). "Least angle regression." *Annals of Statistics*, 32(2), 407–499.
- Bühlmann, P. & van de Geer, S. (2011). *Statistics for High-Dimensional Data*. Springer.

Decision Trees and Random Forests

- Breiman, L., Friedman, J., Olshen, R. & Stone, C. (1984). *Classification and Regression Trees*. Chapman & Hall.

- Breiman, L. (2001). "Random forests." *Machine Learning*, 45(1), 5–32.
- Scornet, E., Biau, G. & Vert, J.-P. (2015). "Consistency of random forests." *Annals of Statistics*, 43(4), 1716–1741.
- Wager, S. & Athey, S. (2018). "Estimation and inference of heterogeneous treatment effects using random forests." *JASA*, 113(523), 1228–1242.
- Belkin, M., Hsu, D., Ma, S. & Mandal, S. (2019). "Reconciling modern machine-learning practice and the classical bias–variance trade-off." *PNAS*, 116(32), 15849–15854.
- Bartlett, P. L., Montanari, A., Rakhlin, A. & Zhou, J. (2020). "Benign overfitting in linear regression." *PNAS*.

Neural Networks and Approximation Theory

- Goodfellow, I., Bengio, Y. & Courville, A. (2016). *Deep Learning*. MIT Press. https://www.deeplearningbook.org
- Barron, A. R. (1993). "Universal approximation bounds for superpositions of a sigmoidal function." *IEEE Trans. Information Theory*, 39(3), 930–945.
- Cybenko, G. (1989). "Approximation by superpositions of a sigmoidal function." *Mathematics of Control, Signals and Systems*, 2(4), 303–314.
- Jacot, A., Gabriel, F. & Hongler, C. (2018). "Neural tangent kernel: convergence and generalization in neural networks." *NeurIPS*.
- Montufar, G., Pascanu, R., Cho, K. & Bengio, Y. (2014). "On the number of linear regions of deep neural networks." *NeurIPS*.
- Rumelhart, D., Hinton, G. & Williams, R. (1986). "Learning representations by back-propagating errors." *Nature*, 323, 533–536.

Variational Inference and Generative Models

- Jordan, M., Ghahramani, Z., Jaakkola, T. & Saul, L. (1999). "An introduction to variational methods for graphical models." *Machine Learning*, 37, 183–233.
- Blei, D., Kucukelbir, A. & McAuliffe, J. (2017). "Variational inference: a review for statisticians." *JASA*, 112(518), 859–877.
- Kingma, D. & Welling, M. (2014). "Auto-encoding variational Bayes." *ICLR*.
- Ho, J., Jain, A. & Abbeel, P. (2020). "Denoising diffusion probabilistic models." *NeurIPS*, 33, 6840–6851.
- Song, Y., Sohl-Dickstein, J., Kingma, D., Kumar, A., Ermon, S. & Poole, B. (2021). "Score-based generative modeling through stochastic differential equations." *ICLR*.
- Lipman, Y., Chen, R., Ben-Hamu, H., Nickel, M. & Le, M. (2023). "Flow matching for generative modeling." *ICLR*.
- Hyvärinen, A. (2005). "Estimation of non-normalized statistical models by score matching." *JMLR*, 6, 695–709.

- Vincent, P. (2011). "A connection between score matching and denoising autoencoders." *Neural Computation*, 23(7), 1661–1674.

Causal Inference

- Pearl, J. (2009). *Causality: Models, Reasoning and Inference*, 2nd ed. Cambridge University Press.
- Rubin, D. B. (1974). "Estimating causal effects of treatments in randomized and nonrandomized studies." *Journal of Educational Psychology*, 66(5), 688–701.
- Angrist, J., Imbens, G. & Rubin, D. (1996). "Identification of causal effects using instrumental variables." *JASA*, 91(434), 444–455.
- Imbens, G. & Rubin, D. (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences*. Cambridge University Press.
- Manski, C. (1995). *Identification Problems in the Social Sciences*. Harvard University Press.
- Spirtes, P., Glymour, C. & Scheines, R. (2000). *Causation, Prediction, and Search*, 2nd ed. MIT Press.
- Angrist, J. (1990). "Lifetime earnings and the Vietnam era draft lottery." *American Economic Review*, 80(3), 313–336.
- Holland, P. (1986). "Statistics and causal inference." *JASA*, 81(396), 945–960.

Double Machine Learning

- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W. & Robins, J. (2018). "Double/debiased machine learning for treatment and structural parameters." *The Econometrics Journal*, 21(1), C1–C68.
- Robinson, P. M. (1988). "Root-N-consistent semiparametric regression." *Econometrica*, 56(4), 931–954.
- Athey, S., Tibshirani, J. & Wager, S. (2019). "Generalized random forests." *Annals of Statistics*, 47(2), 1148–1178.
- Newey, W. K. (1994). "The asymptotic variance of semiparametric estimators." *Econometrica*, 62(6), 1349–1382.
- Kennedy, E. H. (2023). "Semiparametric doubly robust targeted double machine learning: a review." *Statistical Science*.
- Gómez-Bombarelli, R. et al. (2018). "Automatic chemical design using a data-driven continuous representation of molecules." *ACS Central Science*, 4(2), 268–276.